

# Shunting Inhibition and Dendritic Branching Shape Local Credit Assignment

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## Abstract

Biological neurons assign credit across branching dendrites, where synaptic drive, dendritic conductance, local voltage, and somatic teaching signals interact to shape synaptic plasticity. We study conductance-based dendritic networks with E/I synapse banks, shunting inhibition, and tree-structured branch-to-soma coupling, and examine when restricted somatic feedback can approximate compartment-specific backpropagated errors. Exact gradients factor into *local eligibility*  $\times$  *compartment error* terms: the eligibility uses presynaptic activity, driving force, and input resistance, whereas the fast non-local term is a path-specific error obtained by transporting a soma error through dendritic gains. This factorization turns local learning into a credit-signal compression problem. We test the hypothesis that shunting inhibition benefits learning under these constraints when it reshapes the compartment-error field to better match global scalar, per-soma, low-rank, or path-structured feedback. Exact-gradient reconstruction verifies the factorization; path-gain, rank, broadcast-fidelity, inhibition-intervention, and transported-error-oracle diagnostics support the proposed mechanism. Under nonnegative conductances and per-soma 5-factor (5F) feedback, shunting LocalCA remains 5–6 percentage points below matched backpropagation on MNIST, Fashion-MNIST, and figure-ground MNIST, indicating that feedback-field fidelity remains a major bottleneck. These results show how E/I conductance, shunting inhibition, and dendritic branching can reshape credit-signal geometry in restricted local learning.

## 1 Introduction

Unlike point units in standard neural networks, biological neurons assign credit to synapses distributed across branching dendritic trees. Each synapse receives presynaptic drive and samples local membrane voltage, synaptic reversal potential, and total conductance, whose reciprocal is the local input resistance. Inhibitory synapses increase total local conductance, counteracting local excitation while shunting voltage to alter the gain of every dendritic path passing through that compartment.

We examine whether these biophysical ingredients can facilitate local credit assignment (LocalCA). Exact backpropagation [16] assigns a distinct error to every dendritic compartment, whereas a biologically plausible supervisory signal is more likely to arrive as a low-bandwidth somatic or modulatory broadcast [21]. The central question is therefore not whether such a broadcast can reproduce unconstrained backpropagation in every setting, but when the exact dendritic error field is simple enough to be approximated by a low-bandwidth broadcast. Our contribution is not simply to add dendritic structure to a neural network, but to show how conductance-based synapses, shunting inhibition, and dendritic topology determine the geometry of the credit signal.

Starting from conductance-based dendritic voltage equations [1], we derive exact gradients for dendritic trees (Fig. 1; Theorem 1). Each synaptic gradient factorizes into a synapse-local eligibility term and a single non-local compartment error:

$$\text{gradient} = \underbrace{\text{local eligibility}}_{\text{presynaptic drive, } E-V, R^{\text{tot}}} \times \underbrace{\text{compartment error}}_{\text{path-specific non-local term}} .$$

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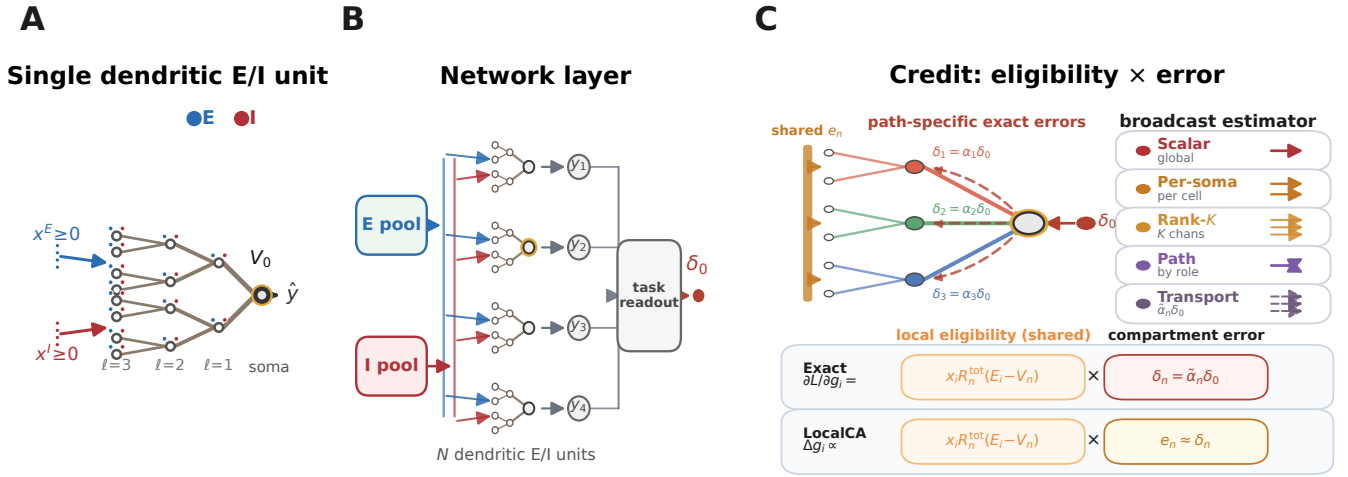


Figure 1: **Model and credit assignment.** (A) Dendritic E/I unit with branch synapse banks. (B) Layer of such units projecting to a task readout;  $\delta_0 = \partial L / \partial V_0$  is the soma/core error used for feedback. (C) Exact updates multiply local eligibility by path-specific errors  $\delta_{u,n}$ ; LocalCA replaces them with broadcast estimates  $e_{u,n}$  of increasing fidelity.

The compartment error is path-specific: under identity upward transfer it is the corresponding soma error multiplied by a conductance-stage path gain through that dendritic tree. LocalCA preserves the exact local eligibility and replaces this path-specific error with a broadcast field.

**What is local in LocalCA?** The fast eligibility variables are synapse- or compartment-local: presynaptic drive, local voltage, driving force, and input resistance. We call the three-factor eligibility-times-error rule 3F, and use 4F and 5F for empirical extensions that multiply the same fast eligibility by slowly estimated branch-level reliability factors. The fast task-dependent non-local quantity is the compartment error, approximated by a broadcast field  $e_n$ . Thus LocalCA variants differ by the broadcast estimator and optional branch-level modulators, while preserving the same local eligibility mechanism.

This factorization turns local credit assignment into a feedback-compatibility problem. An intrinsic low rank of the exact compartment-error matrix is useful, but is not sufficient: the error field must also align with the particular feedback object available to the rule. We therefore distinguish global scalar broadcast, neuron-wise/per-soma broadcast, random low-rank feedback, path-structured feedback, and exact transported-error oracles. Dendritic branching creates the paths; E/I synapse banks set the local conductance state; and shunting inhibition changes the input resistance along each path. Shunting is not assumed to help LocalCA monotonically. It helps restricted feedback when inhibitory conductance attenuates high-gain paths or otherwise makes the exact error field better aligned with the available broadcast.

**Related work.** This work builds on models of dendritic signal integration, branch-level nonlinear computation, divisive normalization, three-factor (3F) plasticity, and biologically plausible teaching signals [35, 1, 3, 4, 36, 9, 12, 8, 18, 19, 20, 27, 28, 29, 23, 46]. It also connects to machine-learning approaches that replace exact backpropagation with local, random, predictive, equilibrium, perturbation, or broadcast feedback signals [16, 15, 14, 17, 32, 22, 33, 34, 30, 31, 37, 38, 39, 40, 41]. Our contribution is complementary: we derive the exact conductance-tree credit object and test when shunting inhibition and dendritic morphology make that object broadcast-compatible.

We test this chain directly. Exact-gradient reconstruction verifies the factorization numerically (Fig. 2). Gradient-fidelity diagnostics are used as diagnostics, not as a claim of backpropagation equivalence: shunting improves directional alignment relative to an additive control, but the absolute cosine remains modest, while scale mismatch is substantially reduced. Exact-error rank and direct broadcast-fidelity diagnostics show that shunting can make the compartment-error field easier to approximate by the feedback field used by the learning rule. Transported-error oracle experiments show that supplying the exact path-transported compartment error substantially reduces the local-learning gap. Together, these results identify feedback fidelity as an important bottleneck in these experiments.

**Contributions.** This paper makes three contributions. First, it derives the conductance-stage path gain, its local-

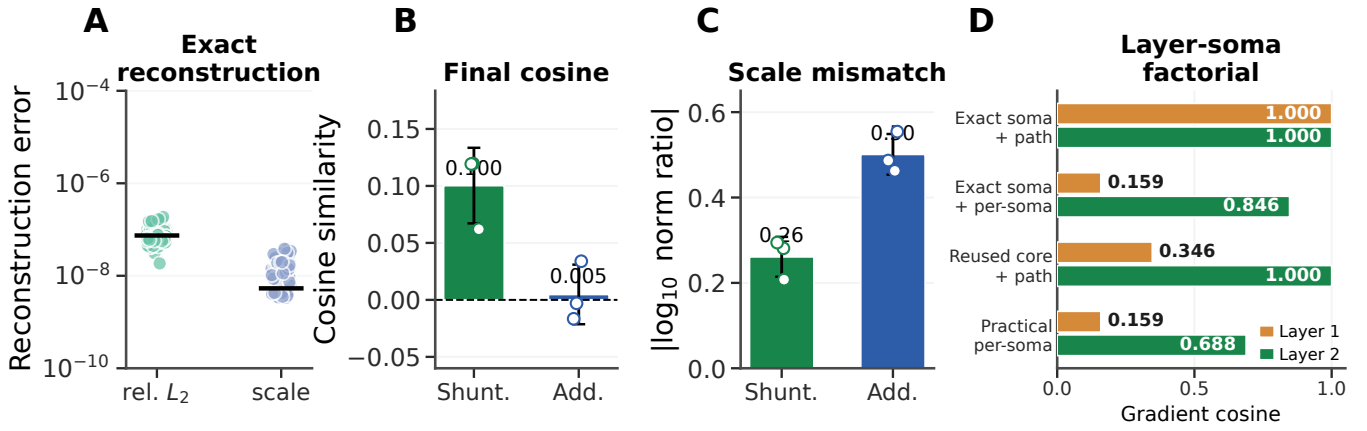


Figure 2: **Exact reconstruction and gradient diagnostics.** (A) Exact-factorization reconstruction matches autograd. (B,C) Final LocalCA-backpropagation cosine and scale mismatch; dots show seeds. (D) Layer-soma factorial diagnostic separating exact vs. reused soma teaching signals and exact path transport vs. per-soma sharing. Metrics/protocol: Appendix A. Error bars in (B,C) are  $\pm 1$  s.d.

eligibility and path-error factorization, and a closed-form inhibitory path-gain ratio. Second, it identifies path-gain and broadcast-compatibility metrics that mediate whether restricted somatic feedback can approximate exact compartment errors. Third, it shows empirically that shunting inhibition can improve this approximation while also identifying regimes where scalar or low-rank feedback fails.

**Scope of evidence.** Analytical statements concern the conductance-tree factorization, path gains, and inhibitory path-gain derivatives under the stated steady-state assumptions. Simulation controls test exact reconstruction, broadcast fidelity, transported-oracle upper bounds, and additive/shunting comparisons. Biological interpretations are restricted to branch-local inhibition and morphology-dependent credit routing; transported-error experiments are oracle diagnostics, not proposed biological learning rules.

## 2 Compartmental Voltage Model and Gradient Derivation

For each dendritic compartment, we use a steady-state conductance model derived from discretized passive cable dynamics [1, 2]. In normalized units, leak and inhibitory reversal potentials are set to 0, the excitatory reversal potential is set to 1, and leak conductance is fixed to 1, so all conductances are expressed relative to leak. Each compartment computes a conductance-weighted voltage: excitatory conductance pulls the voltage toward the excitatory reversal potential, leak and shunting inhibition pull it toward zero, and dendritic coupling transfers child voltages upward toward the soma. This form makes local sensitivity depend on both driving force ( $E - V$ ) and total input resistance  $R^{\text{tot}}$ . Throughout the manuscript we write dendritic morphologies as rooted trees  $[b_1, b_2, \dots, b_D]$ , where each factor gives the fan-in from one dendritic stage to the next on the path to the soma. Thus,  $[3, 3]$  means that each soma aggregates 3 proximal branches and each proximal branch aggregates 3 distal branches, for 9 distal leaves per soma. In the main feedforward experiments, synaptic inputs terminate on dendritic branches rather than directly on the soma. Each branch layer carries explicit excitatory conductance-based synapse banks and, when enabled, learned inhibitory branch-level conductance-based synapse banks driven by nonnegative inputs. This is input-driven shunting inhibition, not a recurrent lateral inhibitory circuit; explicit inhibitory-cell controls are reported in the appendix. The additive and shunting cores are architecture-matched at the tree level and differ only in the branch-level synaptic integration rule. These ingredients play distinct roles in the credit-assignment argument. Branching creates the path structure whereas E/I synapse banks determine the local conductance state. Shunting turns inhibitory conductance into a multiplicative change in path gain through input resistance. The local learning rule then examines whether a low-bandwidth teaching signal can substitute for the exact path-specific error induced by the dendritic tree. **Notation.** A compact reference is provided in Appendix Table S1. We use  $\delta_{0,u}^{(\ell),V} = \partial L / \partial V_{0,u}^{(\ell)}$  for the exact voltage-space error of soma  $u$  in dendritic

layer  $\ell$ ,  $\delta_{0,u}^{(\ell),a} = \partial L / \partial a_{0,u}^{(\ell)}$  for the corresponding activation-space error when soma activity  $a_0 = f_0(V_0)$  is used,  $\delta_0$  for the layer-wide soma/core teaching vector used operationally,  $\bar{\delta}$  for the global scalar obtained by averaging soma coordinates within an example, and  $\delta_{u,n}^{(\ell)} = \partial L / \partial V_{u,n}^{(\ell)}$  for compartment  $n$  of neuron  $u$  in layer  $\ell$ . We use  $\alpha_n^{\text{cond}}$  for the conductance-stage path gain from compartment  $n$  to the soma and  $\tilde{\alpha}_n$  for the effective path gain after including any post-voltage activation derivatives; when upward transfer is identity,  $\tilde{\alpha}_n = \alpha_n^{\text{cond}}$ . We use  $N_E$  and  $N_I$  for the number of excitatory and inhibitory synapses per branch,  $r_n^{4F}$  for the empirical 4F branch-soma covariance proxy,  $\phi_n$  for the 5F bounded branch-level preconditioner,  $\lambda_{\text{mix}}$  for the pathway-broadcast mixing weight, and  $\kappa_j$  for depth-dependent modulation in morphology extensions. All conductances and presynaptic activities in the main experimental setting are nonnegative; additive voltages may be signed comparators, but the conductance variables that define the shunting model are nonnegative.

*Voltage equation and local sensitivities.* Consider compartment  $n$  with synaptic inputs  $j$  (activity  $x_j$ , reversal  $E_j$ , conductance  $g_j^{\text{syn}} \geq 0$ ) and dendritic inputs from children (voltage  $V_j$ , conductance  $g_j^{\text{den}} \geq 0$ ). The steady-state voltage is:

$$V_n = \frac{\sum_j E_j x_j g_j^{\text{syn}} + \sum_j V_j g_j^{\text{den}}}{\underbrace{\sum_j x_j g_j^{\text{syn}} + \sum_j g_j^{\text{den}} + 1}_{g_n^{\text{tot}}}}, \quad R_n^{\text{tot}} = 1/g_n^{\text{tot}}. \quad (1)$$

$V_n$  is a convex combination of reversal potentials, child voltages, and leak. Writing  $\mathcal{S}_n = \{0\} \cup \{E_j\}_j \cup \{V_j\}_j$ , we have  $\min \mathcal{S}_n \leq V_n \leq \max \mathcal{S}_n$  and  $0 < R_n^{\text{tot}} \leq 1$ . The local sensitivities follow directly:

**Proposition 1** (Local Sensitivities).

$$\frac{\partial V_n}{\partial g_i^{\text{syn}}} = x_i R_n^{\text{tot}} (E_i - V_n), \quad \frac{\partial V_n}{\partial V_i} = g_i^{\text{den}} R_n^{\text{tot}}, \quad \frac{\partial V_n}{\partial g_i^{\text{den}}} = R_n^{\text{tot}} (V_i - V_n). \quad (2)$$

Eq. (2) gives the eligibility factors used by the local rules below: presynaptic activity or voltage difference, driving force, and input resistance. If a child transmits post-voltage activity  $a_i = f_i(V_i)$  rather than raw voltage, the dendritic-conductance derivative is  $R_n^{\text{tot}}(a_i - V_n)$  and the child-voltage error includes the local factor  $f'_i(V_i)$ .

*Shunting inhibition as divisive gain control.* An inhibitory synapse with  $E_{\text{inh}} \approx 0$  contributes current  $(0 - V_n)x_j g_j^{\text{syn}}$  and increases  $g_n^{\text{tot}}$ . Its sensitivity is  $\partial V_n / \partial g_j^{\text{syn}} = -x_j R_n^{\text{tot}} V_n$ , which corresponds to multiplicative attenuation (divisive normalization). Shunting is divisive at the voltage level, but its effect on firing rates can be subtractive in some regimes [10]. Inhibitory plasticity can balance excitation dynamically [11]; our learned inhibitory conductances provide a trainable balancing mechanism. The same denominator also gives the direct connection to path gain. Writing  $G_n^E(x)$  and  $G_n^I(x)$  for the total excitatory and inhibitory synaptic conductance impinging on compartment  $n$ ,

$$R_n^{\text{tot}}(x) = (1 + G_n^E(x) + G_n^I(x) + G_n^{\text{den}})^{-1}.$$

Increasing  $G_n^I(x)$  lowers  $R_n^{\text{tot}}(x)$  and therefore multiplicatively suppresses every upstream path whose error must pass through compartment  $n$ . In this sense inhibition can gate credit flow by changing the path gain, even when the inhibitory drive is feedforward rather than lateral. Prop. 2 formalizes this statement after the tree path gain is defined.

*Exact gradients for dendritic trees.* For one neuron/tree in layer  $\ell$ , let  $V_{0,u}^{(\ell)}$  be the somatic voltage and define the exact voltage-space soma error as  $\delta_{0,u}^{(\ell),V} := \partial L / \partial V_{0,u}^{(\ell)}$ . If a decoder receives soma activity  $a_0 = f_0(V_0)$  rather than raw voltage, the decoder supplies an activation-space error  $\delta_0^a$  and the voltage-space boundary is  $\delta_0^V = f'_0(V_0)\delta_0^a$ . The theoretical derivation is stated for the conductance-stage voltage before any post-voltage reactivation. When a branch applies a monotone reactivation before transmitting activity upward, the same recursion holds with additional local derivative factors along the path. All exact-error diagnostics below are computed as pre-reactivation voltage errors, including activation-derivative factors when reactivation is enabled.

**Theorem 1** (Conductance-Stage Backpropagation on a Dendritic Tree). *For a rooted dendritic tree with soma at node 0 and unique parent  $p(n)$  for each non-somatic compartment, suppose the parent receives either the pre-reactivation*

voltage  $V_n$  or a local post-voltage activity  $a_n = f_n(V_n)$ . For identity upward transfer, the loss gradient satisfies

$$\frac{\partial L}{\partial V_n} = \frac{\partial L}{\partial V_{p(n)}} R_{p(n)}^{\text{tot}} g_{n \rightarrow p(n)}^{\text{den}}, \quad (3)$$

with boundary condition  $\partial L / \partial V_0 = \delta_0^V$ . With post-voltage transfer  $a_n = f_n(V_n)$ , the local factor is the product of  $f'_n(V_n)$ ,  $R_{p(n)}^{\text{tot}}$ , and  $g_{n \rightarrow p(n)}^{\text{den}}$ . Because the graph is a tree, each compartment has a unique path to the soma. Defining the conductance-stage path gain

$$\alpha_n^{\text{cond}} = \prod_{(i \rightarrow k) \in \text{path}(n \rightsquigarrow 0)} R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}, \quad \frac{\partial L}{\partial V_n} = \alpha_n^{\text{cond}} \delta_0^V, \quad (4)$$

with  $\alpha_0^{\text{cond}} = 1$  under identity transfer. In the post-voltage case, the exact effective path gain is

$$\tilde{\alpha}_n = \prod_{(i \rightarrow k) \in \text{path}(n \rightsquigarrow 0)} f'_i(V_i) R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}, \quad \frac{\partial L}{\partial V_n} = \tilde{\alpha}_n \delta_0^V, \quad (5)$$

where  $f'_i(V_i)$  is omitted for identity transfer.

*Proof.* Apply the chain rule on the tree-structured computation graph using Prop. 1.  $\square$

The one-step recursion in Eq. (3) and the path product in Eq. (4) separate local conductance transfer from the non-local somatic error for one soma/tree and one layer. Supplying an approximate layer-soma teaching vector, reusing a decoder-derived vector at another layer, or compressing  $\delta_{0,u}^{(\ell),V}$  into a per-soma or global scalar broadcast is an additional feedback-approximation step introduced by LocalCA, not a consequence of the theorem itself.

**Proposition 2** (Inhibitory control of conductance-stage path gain). *Let  $\mathcal{A}(n)$  be the set of compartments on the path from compartment  $n$  to the soma, excluding  $n$  and including the parent compartments through which the error must pass. Holding dendritic coupling conductances fixed, an added inhibitory conductance  $\Delta G_k^I(x) \geq 0$  at any  $k \in \mathcal{A}(n)$  changes the conductance-stage path gain by*

$$\frac{\alpha_n^{\text{cond}}(x; \Delta G^I)}{\alpha_n^{\text{cond}}(x; 0)} = \prod_{k \in \mathcal{A}(n)} \frac{g_k^{\text{tot}}(x)}{g_k^{\text{tot}}(x) + \Delta G_k^I(x)}, \quad \frac{\partial \log \alpha_n^{\text{cond}}}{\partial G_k^I} = \begin{cases} -R_k^{\text{tot}}, & k \in \mathcal{A}(n), \\ 0, & k \notin \mathcal{A}(n). \end{cases} \quad (6)$$

*Proof.* Substitute  $R_k^{\text{tot}} = 1/g_k^{\text{tot}}$  into the path-gain product in Eq. (4). Adding inhibitory conductance changes only the denominator at compartments on the path from  $n$  to the soma, which gives the product ratio and the log derivative in Eq. (6).  $\square$

Prop. 2 makes two directed predictions. First, inhibition is a path gate: it attenuates all credit paths below the inhibited compartment. Second, inhibition is useful for restricted-feedback local learning only when this attenuation makes the distribution of  $\alpha_n^{\text{cond}}$  more uniform or better aligned with the task. If inhibition is too strong, too heterogeneous, or generated by a poorly calibrated inhibitory population, it can suppress useful signals and hurt learning. In the additive control the same inhibitory input changes voltages and local eligibilities, but it does not enter the conductance-stage path recursion as a multiplicative resistance gate.

**When does shunting compress path gains?** The attenuation identity above is not by itself a concentration result. Let  $z_n = \log \alpha_n^{\text{cond}}$  be the log conductance-stage path gain for compartment  $n$ . If added inhibitory conductance contributes a path-dependent attenuation

$$\beta_n(x) = \sum_{k \in \mathcal{A}(n)} \log \left( 1 + \frac{\Delta G_k^I(x)}{g_k^{\text{tot}}(x)} \right), \quad z'_n = z_n - \beta_n, \quad (7)$$

then, over compartments or examples,

$$\text{Var}(z') = \text{Var}(z) + \text{Var}(\beta) - 2\text{Cov}(z, \beta). \quad (8)$$

Thus, shunting narrows the log path-gain field exactly when

$$\text{Cov}(z, \beta) > \frac{1}{2} \text{Var}(\beta). \quad (9)$$

Eq. (7) follows by taking the logarithm of the product ratio in Prop. 2; Eq. (8) then applies the variance identity for  $z - \beta$ . Eq. (9) is a diagnostic condition under one averaging measure, not the empirical mechanism claimed below. In finite trained networks, the covariance margin can disagree with CV and exact-error-rank summaries because these statistics average over different compartment and sample axes. We therefore use direct path-gain dispersion, exact-error compressibility, and inhibition-intervention measurements as the empirical tests, and report the covariance margin as a boundary check in the appendix.

**Corollary 1** (Local–Global Factorization). *The exact synaptic gradient at compartment  $n$  factorizes as:*

$$\frac{\partial L}{\partial g_i^{\text{syn}}} = \underbrace{x_i R_n^{\text{tot}} (E_i - V_n)}_{\text{synapse-local eligibility}} \cdot \underbrace{\frac{\partial L}{\partial V_n}}_{\text{compartment error}}, \quad (10)$$

where the eligibility term depends only on quantities available at synapse  $i$  (presynaptic activity  $x_i$ , input resistance  $R_n^{\text{tot}}$ , driving force  $E_i - V_n$ ), and the compartment error  $\partial L / \partial V_n$  is the sole non-local quantity.

Exact-gradient reconstruction uses the measured  $R_n^{\text{tot}}$  from the forward pass. A biological implementation need not explicitly represent this factor with numerical precision: if branch resistance varies slowly, it can be approximated or absorbed into branch-specific plasticity gain, and it provides a natural site for neuromodulatory regulation of learning rate.

**Proposition 3** (Path-gain dispersion controls 3F alignment). *For a single example before batch averaging, consider one soma/tree with the exact voltage-space soma error  $\delta_0^V$  supplied, and write the local eligibility for conductance parameter  $i$  as  $z_i$ . Let  $n(i)$  be the compartment containing parameter  $i$ . Under identity transfer replace  $\tilde{\alpha}_{n(i)}$  below by  $\alpha_{n(i)}^{\text{cond}}$ ; with post-voltage transfer use the effective gain  $\tilde{\alpha}_{n(i)}$ . Assume both gradient vectors are nonzero, equivalently  $\sum_i w_i > 0$  and  $\sum_i w_i \tilde{\alpha}_{n(i)}^2 > 0$ . The exact and 3F per-soma gradients are*

$$g_i^{\text{BP}} = z_i \tilde{\alpha}_{n(i)} \delta_0^V, \quad \hat{g}_i^{\text{3F}} = z_i \delta_0^V. \quad (11)$$

If transfer derivatives and conductance couplings are nonnegative,  $\tilde{\alpha}_{n(i)} \geq 0$ , so corresponding nonzero components have the same sign. Their cosine is

$$\cos(\hat{g}^{\text{3F}}, g^{\text{BP}}) = \frac{\sum_i w_i \tilde{\alpha}_{n(i)}}{\sqrt{\sum_i w_i} \sqrt{\sum_i w_i \tilde{\alpha}_{n(i)}^2}} = \frac{1}{\sqrt{1 + \text{CV}_w(\tilde{\alpha}_{n(i)})^2}}, \quad w_i = (z_i \delta_0^V)^2. \quad (12)$$

*Proof.* Substitute Eq. (11) into the cosine formula and cancel the common signed eligibility factors through  $w_i$ . The final equality is the weighted identity  $E_w[\tilde{\alpha}^2] = E_w[\tilde{\alpha}]^2 (1 + \text{CV}_w(\tilde{\alpha})^2)$ .  $\square$

Prop. 3 is conditional on the exact layer-soma error. It explains why low effective path-gain dispersion is a useful within-tree credit geometry, while also making clear that practical multi-layer LocalCA must approximate the layer-soma error before this within-tree argument applies. The factorization in Eq. (10) isolates the fast task-error-dependent non-local quantity as  $\partial L / \partial V_n = \alpha_n^{\text{cond}} \delta_0^V$  under identity upward transfer and  $\partial L / \partial V_n = \tilde{\alpha}_n \delta_0^V$  when post-voltage transfer is enabled. The implemented updates in Eqs. (13)–(14) are batch averages of products, so Prop. 3 should be read as the pre-batch geometry of per-example gradient contributions. The 4F and 5F factors introduced below use slower voltage statistics, so they are empirical reliability preconditioners rather than additional error signals. Any practical local rule therefore depends on how well its broadcast signal approximates that exact compartment error, which we evaluate in the gradient-fidelity diagnostic below. In the models studied here, shunting improves this approximation by reducing and stabilizing  $R_n^{\text{tot}}$ , which narrows the spread of local sensitivities across the tree. In the empirical networks, each dendritic branch layer may optionally apply a post-voltage reactivation  $f(V_n)$  before passing activity to the next stage; unless otherwise noted, the manuscript sweeps use a learnable bounded tanh reactivation.

### 3 Local Learning Rules

*Broadcast error approximation.* We approximate the exact compartment error  $\partial L/\partial V_n$  (Corollary 1) with a broadcast signal  $e_n^V$  derived from a soma/core teaching vector. For a linear readout this vector is  $W_{\text{dec}}^\top (\partial L/\partial \hat{y})$ ; for a nonlinear decoder we use the corresponding decoder-input Jacobian product  $J_{\text{dec}}(h_{\text{core}})^\top (\partial L/\partial \hat{y})$  so that feedback is defined in soma/core coordinates. If this vector is used at an earlier matched-width dendritic layer, practical LocalCA assumes neuron-index correspondence and reuses the coordinate as an approximate layer-soma error. This inter-layer teaching approximation is distinct from the within-tree question addressed by Theorem 1: given a layer-soma error, how faithfully can a restricted branch-level field approximate  $\partial L/\partial V_n$ ? The main matched-capacity performance and mechanism runs use per-soma feedback: when soma/core dimension matches a layer’s neuron dimension, each neuron receives one soma-specific teaching coordinate and the same value is shared across that neuron’s dendritic compartments. Conditional on an exact layer-soma error, per-soma sharing isolates the within-tree approximation. Practical multi-layer LocalCA additionally approximates earlier-layer soma errors by reusing final-core coordinates at matched-width layers, as separated explicitly in Fig. 2D. A stricter scalar mode compresses the soma/core vector to one number per example,  $\bar{\delta}_b = d_0^{-1} \sum_{c=1}^{d_0} \delta_{0,bc}^a$ , and broadcasts that number to all neurons and compartments. Thus scalar feedback also discards heterogeneity across soma-error coordinates. For higher-bandwidth controls,  $P_K$  is a fixed random projection from soma-error coordinates to  $K$  feedback channels,  $q_{n,k}$  are fixed compartment-specific mixing coefficients, and  $\Gamma_n$  denotes the pathway-structured transport map for compartment  $n$ . We distinguish five feedback objects: **global scalar broadcast**,  $e_n = \bar{\delta} \mathbf{1}$ ; **per-soma shared broadcast**,  $e_{u,n}$  is one coordinate shared across the compartments of neuron  $u$ ; **rank- $K$  random broadcast**,  $c = P_K \delta_0^a$  and  $e_n = \sum_k q_{n,k} c_k$ ; **path-structured broadcast**,  $e_n = \lambda_{\text{mix}} \bar{\delta} \mathbf{1} + (1 - \lambda_{\text{mix}}) \Gamma_n(\delta_0^a)$  with pathway roles motivated by vectorized dendritic teaching signals [45]; and the **transported oracle**,  $e_n^{\text{transport}} = \tilde{\alpha}_n \delta_0^V = \partial L/\partial V_n$ , an analysis upper bound because it supplies the exact path-transported compartment error from Theorem 1. When reactivation is enabled, exact path transport includes the activation derivatives along the branch-to-soma path. The final local voltage-space error is then obtained by multiplying the branch activation-space error by the local derivative  $f'_n(V_n)$ ; with identity transfer, this reduces to  $\alpha_n^{\text{cond}} \delta_0^V$ . Thus, the main feedforward result tests neuron-wise/per-soma feedback rather than a single layer-wide scalar. Under shunting, this restricted field supports nontrivial learning on the standard classification tasks, while scalar, routed, and rank-bridge controls in the appendix expose regimes where lower-bandwidth feedback collapses branch identity and richer feedback is needed. A local-mismatch heuristic is included only as an appendix control because it performs much worse than the somatic-error broadcasts (Appendix A).

*Three-factor learning rule (3F).*

**Definition 1** (3F Update). *For synaptic and dendritic conductances, with  $a_j = V_j$  under identity upward transfer and  $a_j = f_j(V_j)$  when post-voltage transfer is enabled:*

$$\widehat{\nabla}_{g_j^{\text{syn}}} L = \langle x_j R_n^{\text{tot}} (E_j - V_n) e_n^V \rangle_B, \quad \widehat{\nabla}_{g_j^{\text{den}}} L = \langle R_n^{\text{tot}} (a_j - V_n) e_n^V \rangle_B, \quad (13)$$

where  $\langle \cdot \rangle_B$  denotes the batch average.

The three factors are presynaptic activity  $x_j$  (or a transmitted child activity difference), postsynaptic modulation through driving force and input resistance, and the broadcast voltage-space error  $e_n^V$ . The same rule applies to excitatory and inhibitory synapses. The sign difference comes from the driving force  $(E_j - V_n)$ . We write  $\widehat{\nabla}_g L$  as the local gradient estimate supplied to the optimizer; the optimizer applies the usual learning rate and descent step. Equivalently, one could absorb the sign convention into  $e_n^V$  and treat it as a descent signal.

**Additive control.** Rule (13) is the local gradient of the *shunting* voltage  $V_n = \sum E_j g_j x_j / g_n^{\text{tot}}$ . For the additive E/I control, inhibition enters as a fixed signed voltage contribution rather than through the conductance denominator:

$$V_n^{\text{add}} = \sum_{j \in E} g_j^E x_j^E - \sum_{j \in I} g_j^I x_j^I + \sum_k g_k^{\text{den}} a_k.$$

Equivalently,  $\partial V_n^{\text{add}} / \partial g_j^{\text{syn}} = s_j x_j$  with  $s_j = +1$  for excitatory and  $s_j = -1$  for inhibitory synapses, and  $\partial V_n^{\text{add}} / \partial g_k^{\text{den}} = a_k$ . The additive local gradient therefore has no driving-force or  $R_n^{\text{tot}}$  terms ( $R_n^{\text{tot}} \equiv 1$  by definition since there is no

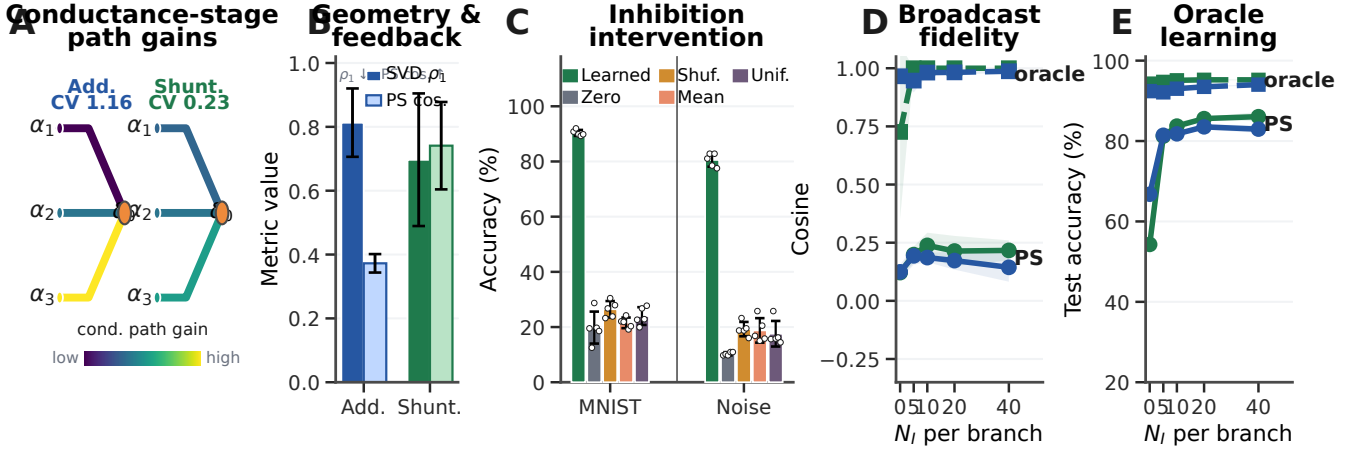


Figure 3: **Mechanistic chain from path gains to learning.** (A) Conductance-stage path-gain field  $\alpha^{\text{cond}}$ . (B) MNIST exact-error geometry and implemented-feedback compatibility: unrestricted rank-1 SVD residual  $\rho_1$  (lower is better) and actual per-soma broadcast cosine (higher is better). (C) Post-training inhibitory-conductance interventions. (D,E) Broadcast fidelity and transported-error learning on noise resilience. CV,  $\rho_1$ , per-soma cosine, and interventions: Appendix A. Error bars are  $\pm 1$  s.d.

denominator). For the identity-transfer additive tree, the exact upward gain is  $\alpha_n^{\text{add}} = \prod_{(i \rightarrow k) \in \text{path}(n \rightarrow 0)} g_{i \rightarrow k}^{\text{den}}$ , with reactivation,  $\tilde{\alpha}_n^{\text{add}} = \prod_{(i \rightarrow k) \in \text{path}(n \rightarrow 0)} f'_i(V_i) g_{i \rightarrow k}^{\text{den}}$ . Thus additive controls have an exact tree gain, but inhibition does not enter that gain through a conductance denominator. Throughout, each architecture uses the learning rule derived from its own forward-pass dynamics. This keeps the comparison architecture-matched rather than applying a shunting-derived rule to an additive model. In implementations with unconstrained raw parameters  $\theta$  and  $g = \text{softplus}(\theta)$ , Eqs. (13)–(14) give conductance-space gradients; before assigning gradients to  $\theta$ , we multiply by  $\partial g / \partial \theta$ . The exact-gradient reconstruction diagnostics include this parameterization factor; Appendix B gives the compact shunting/additive comparison.

*Practical heuristic wrappers: 4F and 5F.* The 4F and 5F variants add slow branch-level preconditioners to the same fast eligibility and broadcast error; they are empirical stabilizers, not additional task-error channels. 4F uses a branch-soma covariance proxy and 5F adds a bounded predictability factor,

$$r_n^{4F} = \frac{\text{Cov}_B(\bar{V}_{n,b}, \bar{V}_{0,b})}{\sqrt{\text{Var}_B(\bar{V}_{n,b}) \text{Var}_B(\bar{V}_{0,b})} + \epsilon}, \quad \phi_n = \frac{\text{Var}(V_n)}{\sigma_{\text{res}}^2 + \epsilon},$$

with implementation clamps  $r_n^{4F} \in [0.1, 2.0]$  and  $\phi_n \in [0.25, 4.0]$ . The appendix specifies the averaging axes, EMA initialization, detaching, and clamp rationale. Empirically, 4F alone gives little improvement over 3F; the practical rule is the 5F update below.

**Definition 2** (5F Update).

$$\widehat{V}_{g_j^{\text{syn}}} L = r_n^{4F} \phi_n \langle x_j R_n^{\text{tot}} (E_j - V_n) e_n^V \rangle_B, \quad \widehat{V}_{g_j^{\text{den}}} L = r_n^{4F} \phi_n \langle R_n^{\text{tot}} (a_j - V_n) e_n^V \rangle_B. \quad (14)$$

Eq. (14) is therefore the main practical LocalCA update; Algorithm 1 (Appendix B) gives the full forward-pass-to-weight-update order. The 3F rule uses the theorem-derived local eligibility with an approximate broadcast error, while 4F and 5F add empirical reliability factors. A feedback-alignment-style random-broadcast argument is included in Appendix C; it is supportive but not central to the main mechanism claim.

## 4 Experiments

*Setup.* The experiments test factorization, gradient fidelity, exact-error geometry, functional dependence on learned inhibition, and transported-error oracles using architecture-matched additive/shunting cores and local/backprop

training on MNIST [43], Fashion-MNIST [42], figure-ground MNIST, and noise resilience; CIFAR-10 [44], cue routing, DFA, and detailed protocols are in the appendix. Unless stated otherwise, local runs use 5F with per-soma feedback, conductance runs use nonnegative first-layer drive, and main mechanism/oracle/performance results use five seeds. Transported error is an oracle upper bound from Theorem 1, not the proposed biological rule. Figure shorthand: BP backpropagation, PS per-soma feedback, PT exact path transport, PV pathway-vector feedback, Or. transported oracle,  $K$  random low-rank channels,  $N_I$  inhibitory synapses/branch, pp percentage points, and MN/FM/FG = MNIST/Fashion-MNIST/figure-ground MNIST. Metrics, implementation, and tasks are defined in Appendices A, B, and D.

*From exact factorization to credit-signal fidelity.* We compare local and backprop gradients on matched batches and parameters using parameter-count-weighted cosine and scale mismatch  $|\log_{10}(\|g_p^{\text{local}}\|/\|g_p^{\text{bp}}\|)|$ . Shunting improves the final diagnostic, though absolute directional agreement remains weak: cosine rises from  $0.005 \pm 0.026$  to  $0.100 \pm 0.033$ , and scale mismatch falls from  $0.50 \pm 0.05$  to  $0.26 \pm 0.05$ . Exact-gradient reconstruction matches autograd to numerical precision (Fig. 2A). The layer-soma factorial diagnostic separates the two practical approximations: exact layer-soma errors plus path transport recover autograd, while per-soma compression and reused final-core coordinates mainly damage the first layer (Fig. 2D). Thus the central non-local approximation is the broadcast substitute for  $\partial L/\partial V_n$ , not the tree recursion.

*Mechanistic chain: path gains, broadcast fidelity, and oracle transport.* Figure 3 gives the central empirical chain. At matched inhibition ( $N_I=5$ ), shunting narrows the MNIST conductance-stage path-gain component (CV 0.23 vs. 1.16 for additive), lowers the unrestricted rank-1 residual ( $0.70 \pm 0.21$  vs.  $0.81 \pm 0.11$ ), and improves compatibility with the implemented feedback field (actual per-soma broadcast cosine  $0.74 \pm 0.13$  vs.  $0.37 \pm 0.18$ ). Panel A reports the conductance-stage component  $\alpha^{\text{cond}}$  and is an architectural path-gain diagnostic rather than a direct validation of Prop. 3; the appendix defines the corresponding effective-gain diagnostics when reactivation is enabled. The SVD and broadcast-cosine bars test intrinsic geometry and implemented-feedback compatibility, respectively.

Learned branch inhibition is functionally important: zeroing, shuffling, or clamping it drops MNIST from about 90% to 20–26% and noise resilience from about 81% to 10–19%. Direct fidelity and oracle panels probe the same bottleneck: under shunting, per-soma feedback better tracks exact compartment errors, transported-error fidelity approaches 0.95–1.0, and transported learning lifts both architectures to 92–95% on noise resilience.

*Matched-capacity performance.* In the matched-capacity regime, shunting LocalCA remains within 5–6 percentage points of matched backpropagation on MNIST, Fashion-MNIST, and figure-ground MNIST (Table S4; Fig. 4). Controls in Fig. 4D identify a feedback-field bottleneck: exact path transport reaches the matched shunting backprop reference, activation choice has little effect, and additive gain/normalization variants do not rescue the additive baseline. Fig. 5 shows that 5F is the strongest practical LocalCA variant, soma-derived feedback is essential, and higher-rank/path feedback improves the noise-resilience bridge.

*Regime dependence of the shunting advantage.* With matched per-soma feedback, shunting is modestly better on MNIST, Fashion-MNIST, and figure-ground MNIST, and the gap grows on inhibition-sensitive noise resilience. Morphology shifts the useful inhibitory regime, so the shunting advantage is not monotonic in  $N_I$  or tree depth (Fig. 4B,C; Appendix Fig. S6).

## 5 Discussion

We studied local credit assignment in conductance-based dendritic networks with explicit E/I conductance synapses, shunting inhibition, and tree-structured branching. Exact gradients separate synapse-local eligibility–presynaptic activity, driving force, and input resistance—from one path-specific compartment error, making LocalCA an error-approximation problem. The experiments support a coherent mechanism: branching creates credit paths, E/I synapse banks set the conductance state, and shunting modulates input resistance along those paths. In the regimes studied here this lowers exact-error rank, improves feedback-fidelity diagnostics, reduces gradient-scale mismatch, and lets transported-error oracles substantially reduce the local-learning gap. Post-training interventions further show that learned inhibitory conductance is sample-dependent and functionally important to the trained forward computation.

The operating regime matters. Shunting is not universally beneficial, more inhibition is not always better, and the practical 5F rule is an empirical bounded preconditioner. Exact decoder-to-soma feedback requires the decoder

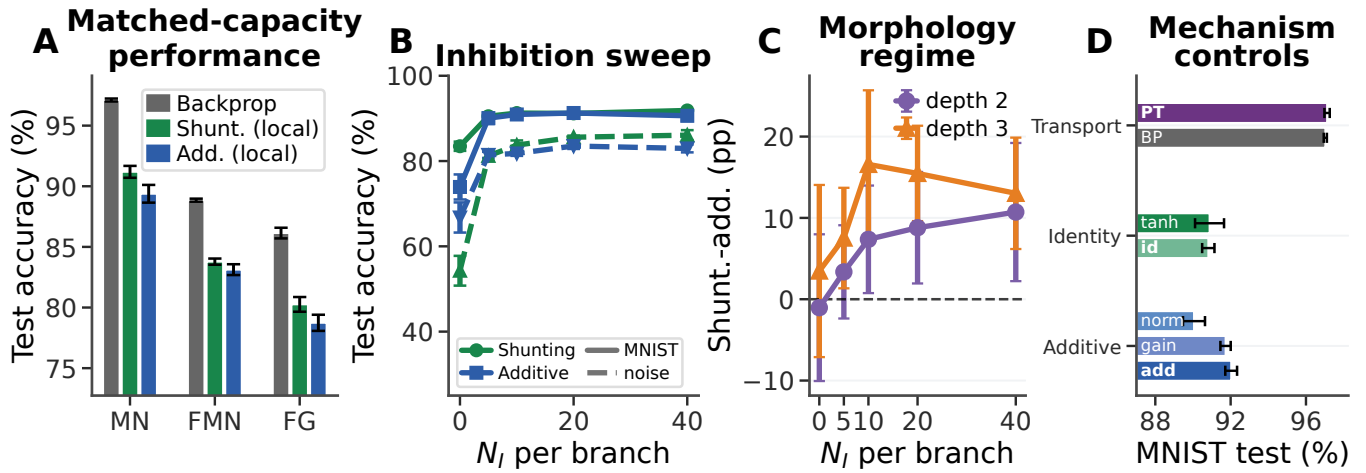


Figure 4: **Matched-capacity performance and regime dependence.** (A) Matched shunting backpropagation reference vs. per-soma 5F LocalCA; additive backpropagation references are reported in Table S4 where matched five-seed runs are available. (B) Inhibitory dose-response. (C) Morphology-dependent shunting advantage. (D) MNIST controls: BP, exact path transport (PT), activation choice, and additive gain/normalization variants. Values/protocols: Table S4; Appendix A. Error bars denote  $\pm 1$  s.d.

Jacobian or linear readout to supply the correct soma/core coordinate; practical multi-layer LocalCA also approximates layer-soma errors by reusing coordinates at earlier matched-width layers. The core theorem covers conductance-stage branch parameters under steady-state, non-spiking dynamics; reactivation, decoder, router, top- $k$ , 4F/5F batch-time statistics, and HSIC terms are implementation-specific updates. Learned synapse selection may also reshape the eligibility basis so a fixed or low-rank feedback field aligns better with active synaptic drive, analogous in spirit to DFA alignment, but we treat this as a hypothesis for future diagnostics rather than a demonstrated mechanism here. Eq. (9) is interpretive, and activation shape, morphology, task structure, and feedback rank can change broadcast compatibility. The inhibition studied here is input-driven rather than recurrent lateral inhibition, and post-training interventions establish forward necessity rather than isolating credit-assignment causality. We therefore claim a credit-geometry mechanism supported by exact reconstruction, fidelity diagnostics, controls, and oracle transport, not competitive CIFAR-10 performance or sufficiency of a single global scalar broadcast for routed/pathway-specific tasks.

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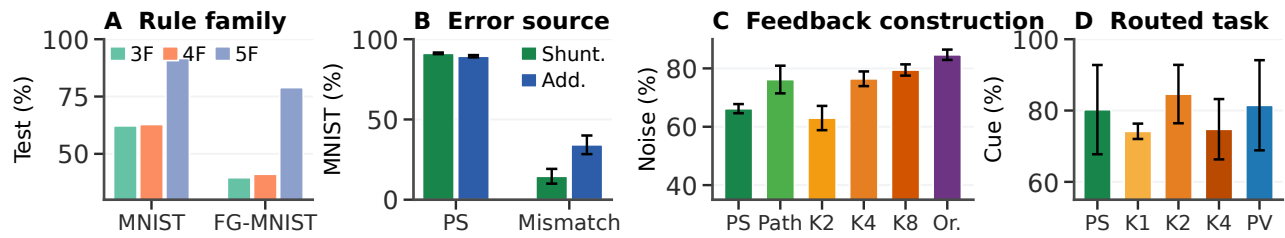


Figure 5: **Rule and feedback controls.** (A) Rule ablation. (B) Error-source ablation. (C) Noise-resilience feedback ladder: PS, path proxy, low-rank  $K$ , and transported oracle. (D) Cue-routing rank diagnostic with pathway-vector feedback (PV). Details: Appendix Fig. S1; Table S8.

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## A Supplementary Empirical Results

This appendix records the supporting diagnostics, implementation details, and task protocols needed to interpret the paper’s main claims. It is organized as a reading path rather than as a chronological log: this section contains empirical diagnostics and boundary cases; Appendix B gives the implemented model and optimizer details; Appendix C collects theory-adjacent extensions; and Appendix D defines benchmark and synthetic-task protocols.

**Scope notes.** Several controls are intentionally negative or mixed. The log-gain covariance margin from Eq. (9) is not used as a positive main result, and the pathway-vector feedback variant does not beat the best unstructured low-rank cue-routing control. These diagnostics delimit the claim rather than smoothing over failure modes.

| Symbol                                             | Name                                | Meaning in this manuscript                                                                                                     |
|----------------------------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| $V_n, a_n$                                         | Compartment voltage/activity        | Conductance-stage voltage and optional post-voltage activity $a_n = f_n(V_n)$ for compartment $n$ .                            |
| $g^{\text{syn}}, g^{\text{den}}$                   | Synaptic and dendritic conductances | Nonnegative conductance parameters for branch synapses and child-to-parent coupling.                                           |
| $R_n^{\text{tot}}$                                 | Input resistance                    | Reciprocal of total conductance at compartment $n$ ; the implementation uses the stabilized $R_{n,\varepsilon}^{\text{tot}}$ . |
| $E_j - V_n$                                        | Driving force                       | Difference between synaptic reversal potential and local voltage; multiplies presynaptic drive in conductance eligibility.     |
| $\delta_{0,u}^{(\ell),V}, \delta_{0,u}^{(\ell),a}$ | Soma voltage/activity error         | Exact layer-soma error for neuron $u$ in layer $\ell$ , before or after post-voltage activation.                               |
| $\delta_{u,n}^{(\ell)}$                            | Compartment error                   | Exact pre-reactivation voltage-space error for compartment $n$ of neuron $u$ in layer $\ell$ .                                 |
| $\delta_0, \bar{\delta}$                           | Broadcast source coordinates        | Soma/core teaching vector and its global scalar average used by restricted-feedback controls.                                  |
| $\alpha_n^{\text{cond}}, \tilde{\alpha}_n$         | Path gain                           | Conductance-stage and activation-derivative-corrected gain transporting soma error to compartment $n$ .                        |
| $e_n^V$                                            | Broadcast voltage-space error       | LocalCA approximation to the exact compartment error used in the 3F–5F updates.                                                |
| $r_n^{\text{4F}}, \phi_n$                          | Branch-level preconditioners        | Slowly estimated 4F covariance proxy and 5F bounded predictability factor.                                                     |
| $N_E, N_I$                                         | Synapse counts                      | Excitatory and inhibitory synapses per dendritic branch.                                                                       |

Table S1: **Notation reference.** Main symbols used in the derivation, local-learning rules, and diagnostics.

## A.1 Reproducibility Statement

Reproducibility is documented at a protocol level in this manuscript. The accompanying source package contains the core model and training code, diagnostic utilities, representative configuration files, tests, figure-source summaries, and analysis scripts used for the manuscript. Table S2 summarizes the experiment families, seed protocols, and checkpoint-selection rules. Code will be released publicly at the time of arXiv posting with an immutable repository tag, pinned environment file, end-to-end launch scripts, and expected summary checks.

| Experiment family                                           | Architecture / condition                                                                              | Seeds                                                                            | Selection / reporting                                                                   |
|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Exact-factorization and gradient diagnostics                | Nonnegative-input E/I dendritic cores, [128, 128] soma layers, [3, 3] tree, 3F–5F local rules         | Five seeds for main summaries; three seeds where indicated for gradient dynamics | Diagnostic batches or fixed checkpoints, reported as gradient/factorization diagnostics |
| Layer-soma factorial diagnostic                             | Two-layer shunting checkpoints with exact/reused soma errors and transported/per-soma branch fields   | Six fixed checkpoints                                                            | Fixed-checkpoint diagnostic; no validation selection                                    |
| Path-gain, exact-error rank, and inhibition intervention    | Matched additive and shunting cores at fixed inhibition; post-training intervention states            | Five seeds per main condition                                                    | Trained checkpoints; post-training interventions are evaluated on held-out data         |
| Transported-error oracle and feedback-construction controls | Per-soma, random low-rank, path-structured, and exact transported-error feedback                      | Five seeds for main oracle and feedback-ladder summaries                         | Validation-selected checkpoints; held-out test summaries                                |
| Matched-capacity performance                                | MNIST, Fashion-MNIST, figure-ground MNIST; per-soma 5F LocalCA and matched backpropagation references | Five seeds                                                                       | Validation-selected configurations; held-out test accuracy                              |

Table S2: **Protocol-level reproducibility summary.** The table records the experiment families, model settings, seed counts, and selection rules used in the manuscript. Dependency specifications are provided through `requirements.txt`, `pyproject.toml`, and `setup/environment.yml`; exact end-to-end commands and summary hashes will be provided with the public repository tag.

## A.2 Diagnostic Metrics

The gradient-fidelity diagnostics compare LocalCA and backpropagation gradients at matched parameters and batches. For a parameter block  $p$ , let  $g_p^{\text{local}}$  and  $g_p^{\text{bp}}$  be the two gradients and let  $w_p = \text{numel}(p) / \sum_q \text{numel}(q)$ . The reported weighted cosine is

$$C_w = \sum_p w_p \frac{\langle g_p^{\text{local}}, g_p^{\text{bp}} \rangle}{\|g_p^{\text{local}}\|_2 \|g_p^{\text{bp}}\|_2 + \epsilon},$$

and weighted scale mismatch is

$$S_w = \sum_p w_p \left| \log_{10} \frac{\|g_p^{\text{local}}\|_2 + \epsilon}{\|g_p^{\text{bp}}\|_2 + \epsilon} \right|.$$

For exact-error compressibility, we form a matrix  $E$  whose rows are examples and whose columns are compartment–neuron coordinates of the exact pre-reactivation compartment error. If  $E = U\Sigma V^\top$  and  $\sigma_i$  are the singular values, the unrestricted rank-1 residual and participation rank are

$$\rho_1 = \frac{\sqrt{\sum_{i>1} \sigma_i^2}}{\sqrt{\sum_i \sigma_i^2}}, \quad r_{\text{part}} = \frac{(\sum_i \sigma_i^2)^2}{\sum_i \sigma_i^4}.$$

This is an intrinsic geometry metric: both the sample coefficients and spatial template are chosen in hindsight by SVD. The constrained shared-field residual instead fixes the spatial template to all ones and allows only the best per-example scalar,

$$\rho_1 = \frac{\|E - \text{rowmean}(E)\mathbf{1}^\top\|_F}{\|E\|_F}.$$

The per-soma/blockwise residual matches the practical feedback object more closely. Let  $\Pi_{\text{PS}}$  replace, for each sample and soma, all compartments of that soma by their optimal shared coefficient. Then

$$\rho_{\text{PS}} = \frac{\|E - \Pi_{\text{PS}}E\|_F}{\|E\|_F}.$$

The broadcast-conditioned residual uses the actual feedback field  $B$  generated by the implemented mode,

$$\rho_{\text{bc}} = \frac{\|E - B\|_F}{\|E\|_F},$$

with a corresponding flattened cosine and, where useful, a globally rescaled version of  $B$ . These constrained metrics are emitted by `scripts/measure_error_rank_diagnostics.py` alongside the unrestricted SVD metrics, so intrinsic error geometry and compatibility with the implemented broadcast field can be reported separately. Path-gain dispersion is the coefficient of variation of  $\alpha_n^{\text{cond}}$  or  $\tilde{\alpha}_n$  over the measured sample and compartment field, depending on whether reactivation is enabled. This unweighted architectural statistic differs from the eligibility-weighted  $\text{CV}_w(\tilde{\alpha}_{n(i)})$  in Prop. 3. Compartment-error fidelity is the cosine between the tested broadcast field and the exact compartment-error field after flattening over the same sample and compartment axes.

### A.3 Rule and Feedback Controls

These controls separate three questions that are deliberately collapsed in the main matched-capacity performance summary: which local-rule wrapper is practical, whether the somatic error source matters, and how performance changes as the feedback field gains bandwidth or path information.

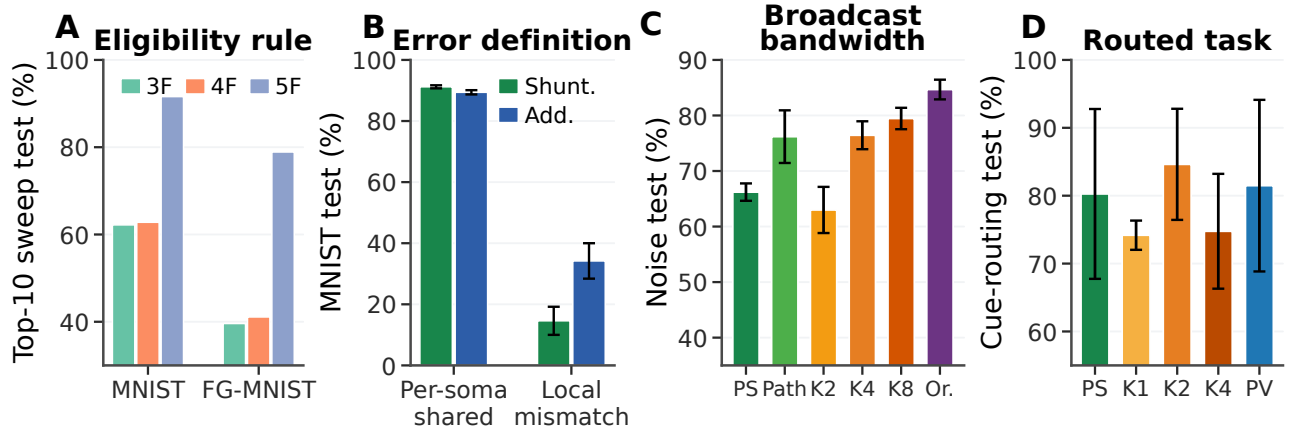


Figure S1: **Detailed rule and feedback controls.** (A) Eligibility-rule ablation: 3F, 4F, and 5F. (B) Error-source ablation: per-soma/shared feedback vs. local mismatch. (C) Feedback-construction ladder on noise resilience: PS per-soma/shared, recursive Path, random low-rank K2/K4/K8, and transported Oracle. (D) Cue routing: PS per-soma/shared, random low-rank K1/K2/K4, and pathway-vector feedback.

### A.4 Rule-Family Ranking

Table S3 keeps the 3F/4F/5F optimizer distinction explicit: the theorem-derived eligibility is central to the proposed mechanism, but the reported LocalCA performance results use the empirical 5F stabilizer.

### A.5 5F Stabilizer Sensitivity

Because the main LocalCA runs use the empirical 5F stabilizer, we also tested whether performance depends sharply on the preconditioner clamp or the EMA rate used to estimate branch statistics. Across three-seed MNIST sensitivity checks, tightening or widening the clamp and varying the EMA rate leaves test accuracy near the default setting.

| Dataset             | Rule | Mean valid, 10 validation-selected configs | Mean test, same 10 configs |
|---------------------|------|--------------------------------------------|----------------------------|
| MNIST               | 3F   | 0.611                                      | 0.622                      |
| MNIST               | 4F   | 0.620                                      | 0.628                      |
| MNIST               | 5F   | <b>0.912</b>                               | <b>0.916</b>               |
| Figure-ground MNIST | 3F   | 0.398                                      | 0.396                      |
| Figure-ground MNIST | 4F   | 0.411                                      | 0.411                      |
| Figure-ground MNIST | 5F   | <b>0.807</b>                               | <b>0.789</b>               |

Table S3: **Rule-family ranking.** Mean over the ten best validation-selected configurations from local-performance sweeps; this is not top-10 classification accuracy, and test scores are reported only after validation selection.

This does not make 5F part of the exact factorization; it supports interpreting it as a bounded branch-level reliability preconditioner rather than a brittle tuned error source.

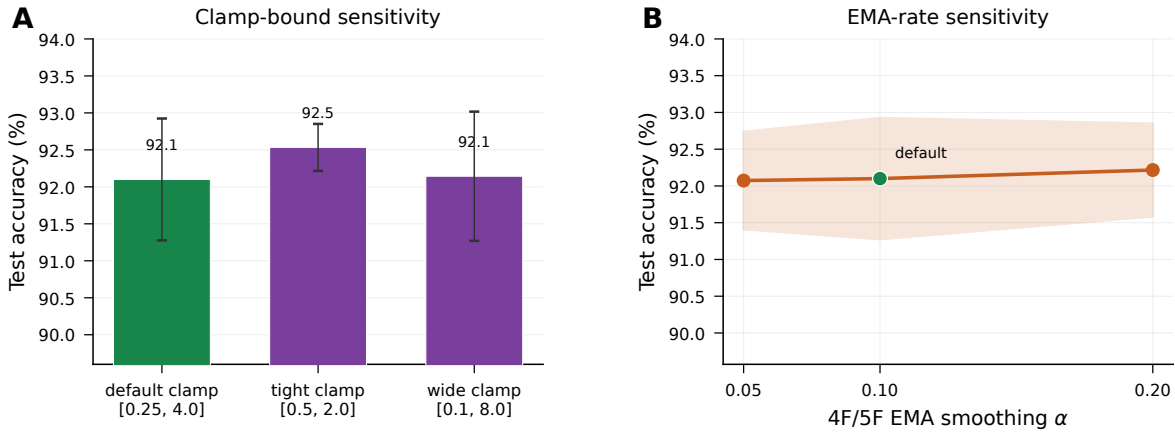


Figure S2: **5F stabilizer sensitivity.** (A) Changing the 5F preconditioner clamp from the default [0.25, 4.0] to tighter [0.5, 2.0] or wider [0.1, 8.0] bounds leaves MNIST test accuracy near 92%. (B) Changing the 4F/5F EMA smoothing rate over  $\alpha \in \{0.05, 0.10, 0.20\}$  gives similarly stable performance. Error bars and bands are  $\pm 1$  s.d. across 3 seeds.

## A.6 Local Performance and Regime Dependence

Table S4 reports the matched-capacity performance numbers used in the main text. The point of this table is not to claim state-of-the-art benchmark performance, but to show the remaining LocalCA-to-backprop gap under matched dendritic capacity and per-soma 5F feedback.

## A.7 Extended Gradient Analysis and $N_I$ Sweep Detail

Figure 2 in the main text summarizes final-state gradient diagnostics and the layer-soma factorial check. The supplementary view below (Fig. S3) separates static scale checks, inhibitory dose-response curves, and individual Fashion-MNIST seed behavior so that the aggregate claims are not driven by a single summary panel.

| Dataset             | Shunt. BP | Add. BP           | Shunt. local      | Add. local        | Shunt. gap (pp) |
|---------------------|-----------|-------------------|-------------------|-------------------|-----------------|
| MNIST               | 0.971     | 0.970 $\pm$ 0.001 | 0.911 $\pm$ 0.005 | 0.894 $\pm$ 0.007 | 6.0             |
| Fashion-MNIST       | 0.889     | 0.876 $\pm$ 0.001 | 0.838 $\pm$ 0.003 | 0.831 $\pm$ 0.004 | 5.1             |
| Figure-ground MNIST | 0.861     | –                 | 0.803 $\pm$ 0.006 | 0.787 $\pm$ 0.007 | 5.8             |

Table S4: **Local performance.** Backpropagation references are matched-capacity standard-training runs. Additive backpropagation is reported where a matched five-seed reference is available; the figure-ground additive backpropagation reference was not available in the matched five-seed performance outputs. Local values are 5F with per-soma feedback and local decoder updates. The figure-ground LocalCA rows use the HSIC auxiliary objective (weight 0.01); the shunting BP reference is the standard cross-entropy matched-capacity reference. Error bars are  $\pm 1$  s.d. across 5 seeds where available; gaps are percentage points.

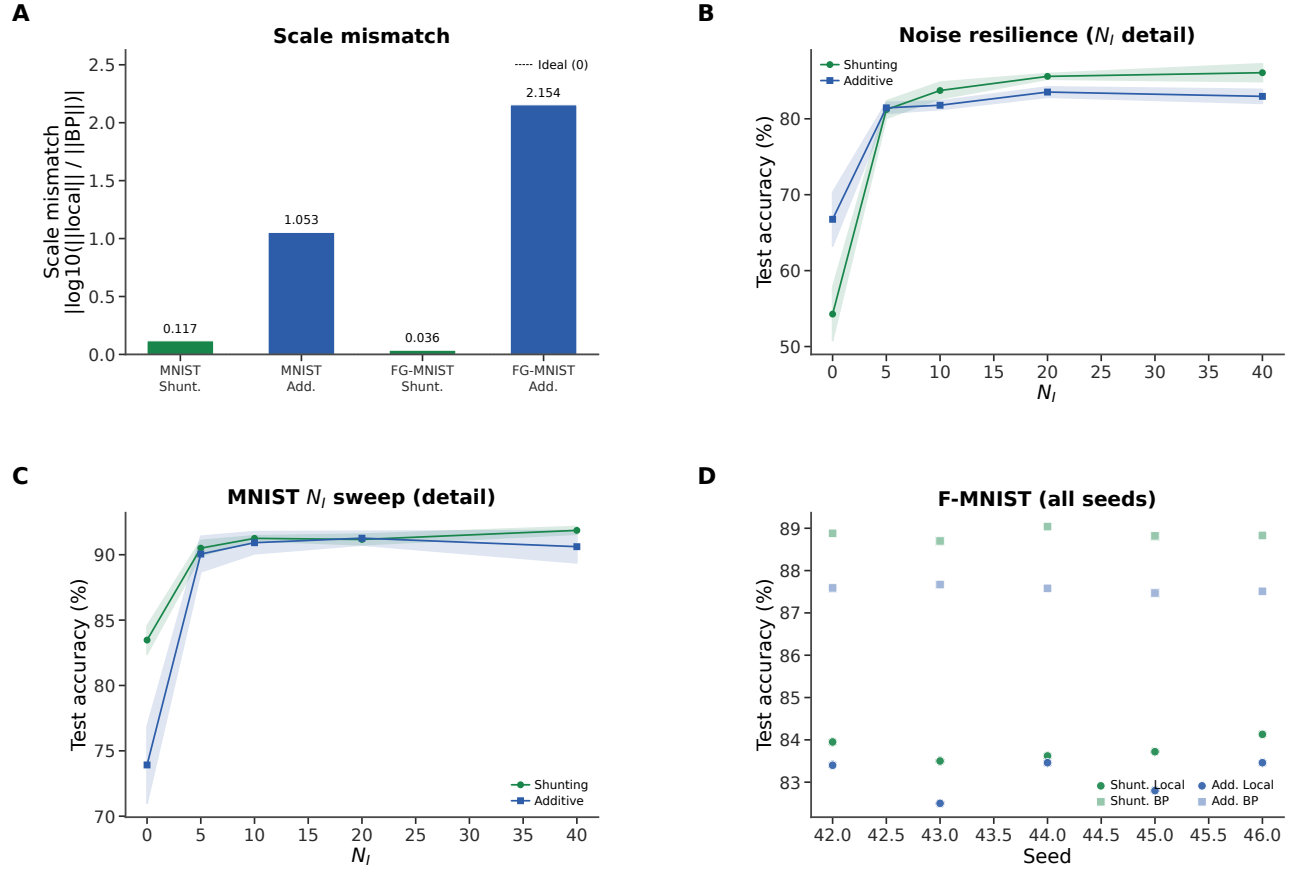


Figure S3: **Extended gradient and  $N_I$  analysis (supplementary).** (A) Log-scale mismatch,  $|\log_{10}(\|g^{\text{local}}\|/\|g^{\text{BP}}\|)|$ , from a separate matched-weight static diagnostic; these values should not be compared directly to the checkpointed final-state diagnostic in Fig. 2. The ideal value is 0; shunting is near ideal (0.117), whereas additive shows order-of-magnitude distortion ( $> 1.0$ ). (B) Noise-resilience  $N_I$  dose-response with error bands ( $\pm 1$  s.d.). (C) MNIST  $N_I$  dose-response detail with error bands. (D) Fashion-MNIST individual seeds for all conditions, showing consistency across runs.

**Alignment dynamics are not a zero-gradient artifact.** To check whether low additive alignment could be caused by vanishing gradients, we analyzed a separate 3-seed checkpoint trajectory with parameter-level LocalCA and backprop gradients. The additive local gradients are nonzero throughout training (weighted norm range  $9.5 \times 10^{-4}$ – $9.5 \times 10^{-3}$ ), and the corresponding backprop gradients are also nonzero ( $2.3 \times 10^{-5}$ – $5.7 \times 10^{-3}$ ). The low additive cosine therefore reflects directional and scale mismatch, not absent gradients: additive 5F cosine stays near zero from epoch 0 to epoch 50 (0.018 to 0.005), while its local/backprop norm ratio falls from about  $8.8 \times 10^2$  to 0.69 (Fig. S4).

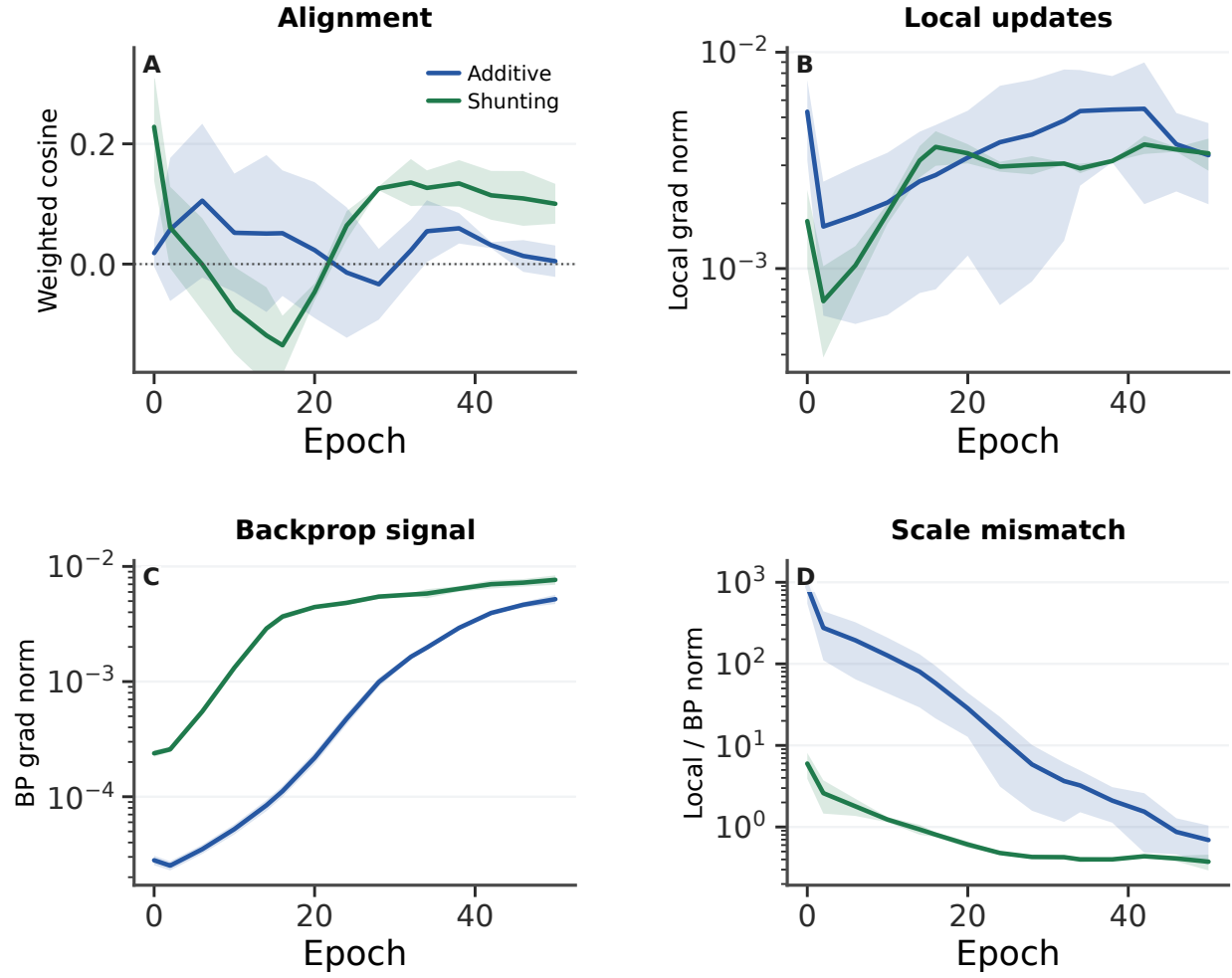


Figure S4: **Alignment dynamics with explicit gradient-norm checks.** (A) Weighted cosine similarity between LocalCA and backpropagation gradients over training for the 5F rule. (B,C) Weighted local and backprop gradient norms stay nonzero for both additive and shunting models. (D) The additive control begins with a much larger scale mismatch, then approaches the backprop scale without recovering strong directional alignment. Shading is s.d. across 3 seeds.

## A.8 Verification and Seed Robustness

These checks (Fig. S5) compare the main seed set with held-out seeds and report the HSIC-weight sensitivity for figure-ground MNIST. The additional matched experiments use the same objective-weight comparison on both seed sets, so seed robustness and HSIC contribution can be summarized as a matched two-factor control.

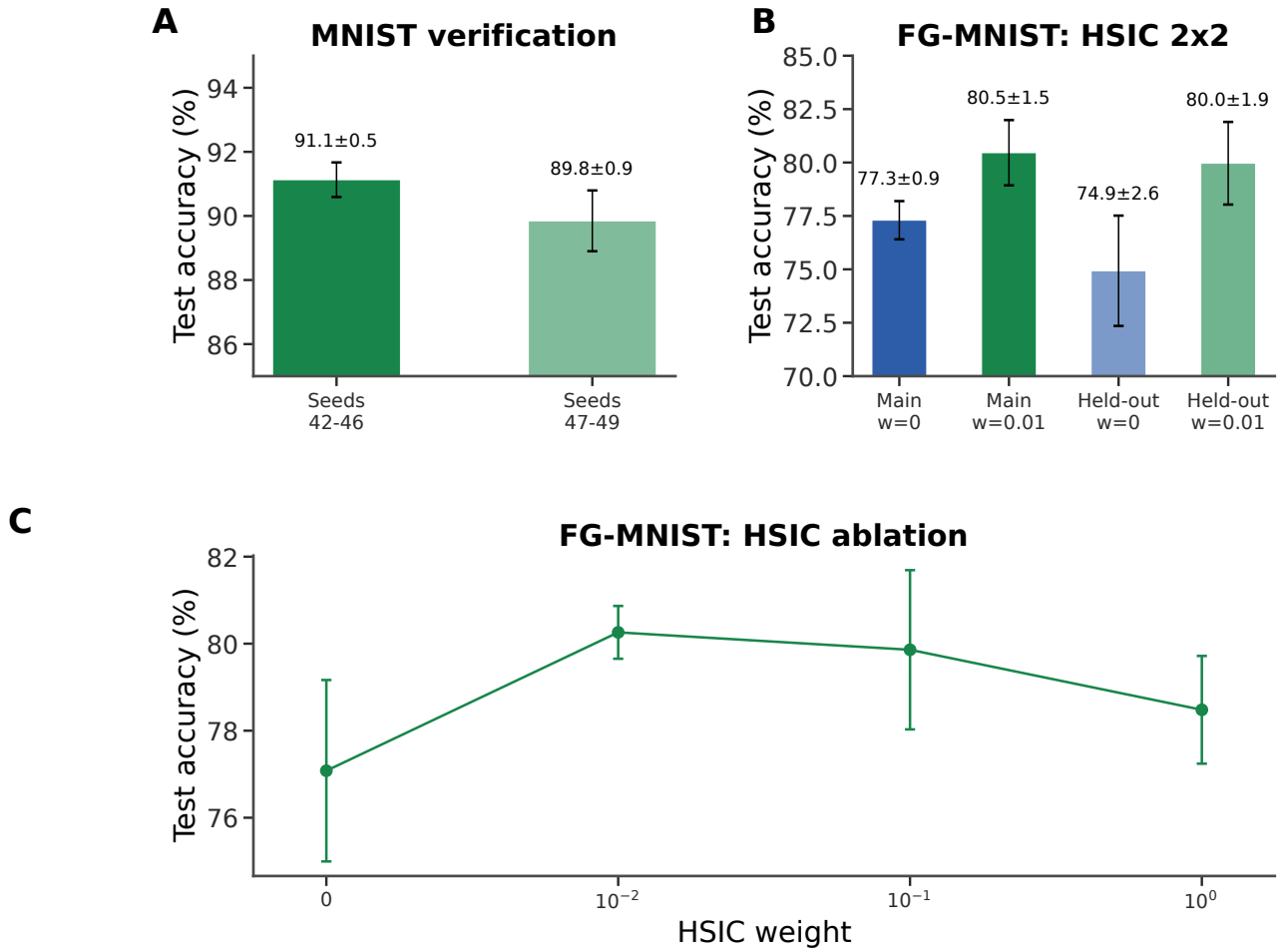


Figure S5: **Verification and seed robustness (supplementary).** (A) MNIST verification: repeat runs with the submitted code on main seeds (42–46) yield  $91.1 \pm 0.5\%$ ; held-out seeds (47–49) yield  $89.8 \pm 0.9\%$ , showing robustness across random seeds. (B) Matched figure-ground MNIST HSIC control: both main seeds (42–46) and held-out seeds (47–49) compare HSIC weight 0 against 0.01. (C) HSIC weight ablation on figure-ground MNIST: moderate weights (0.01–0.1) perform best.

## A.9 Additional Stress Tests

These stress tests probe two ways in which low-bandwidth broadcast can become unreliable: deeper dendritic paths and noisy teaching signals. They are not separate main benchmarks; they delimit where shunting remains useful and where additive local learning degrades.

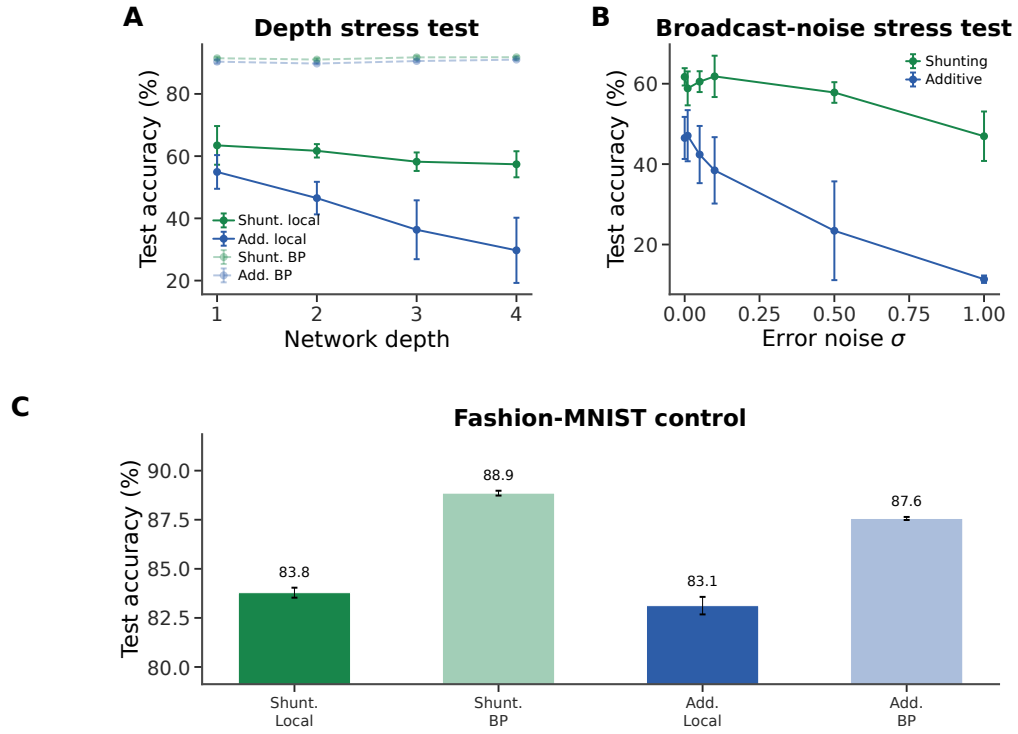


Figure S6: **Additional stress tests.** (A) Depth scaling: shunting local (green) degrades more gracefully from 63.5% to 57.4%; additive local (blue) drops from 54.9% to 29.7%. Matched backpropagation references (dashed) remain stable. (B) Broadcast-noise robustness: shunting remains stable for  $\sigma \leq 0.1$ ; additive degrades rapidly and reaches chance at  $\sigma = 1$ . (C) Fashion-MNIST: the shunting gain is modest on this cleaner benchmark (83.8% local vs. 88.9% backprop), consistent with the regime-dependent interpretation in the main text.

## A.10 FA/DFA Baseline Comparison

Figure S7 includes feedback alignment as an adjacent local-learning reference point. DFA can be evaluated on the dendritic cores, whereas the FA condition is marked unavailable for the block-structured dendritic updates because the required random feedback matrices are not dimensionally compatible with the branch-level parameterization.

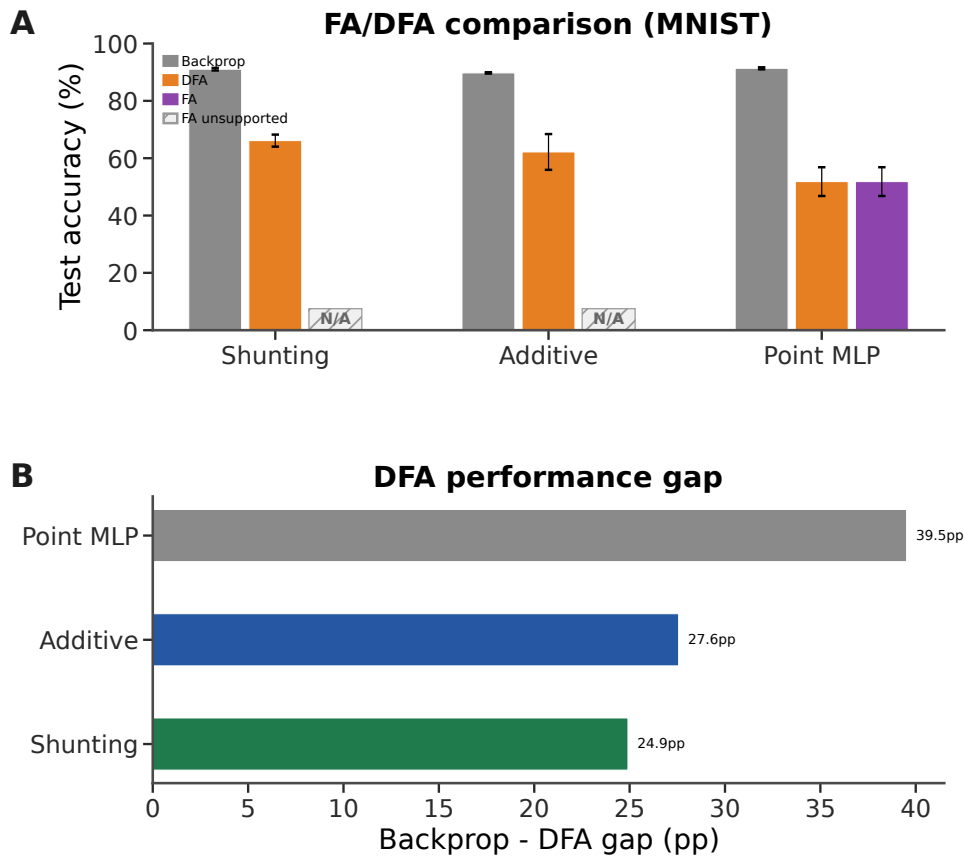


Figure S7: **Feedback alignment baselines (MNIST)**. **(A)** Grouped comparison: standard backprop (neutral gray), DFA (orange), and FA (purple). Hatched N/A markers indicate that FA is not implemented for the block-structured dendritic architectures because the random feedback matrices are not dimensionally compatible with the tree-structured branch updates. DFA achieves 66.1% on shunting vs. 62.2% additive vs. 51.8% point MLP. **(B)** Backprop–DFA gap: dendritic architectures (24.9–27.6 pp) show smaller gaps than point MLPs (39.5 pp), suggesting conductance-based architecture is partially compatible with random feedback.

### A.11 CIFAR-10 Results

Figure S8 and Table S5 report a CIFAR-10 control ladder, used as a harder-dataset mechanistic stress test rather than a competitive vision benchmark.

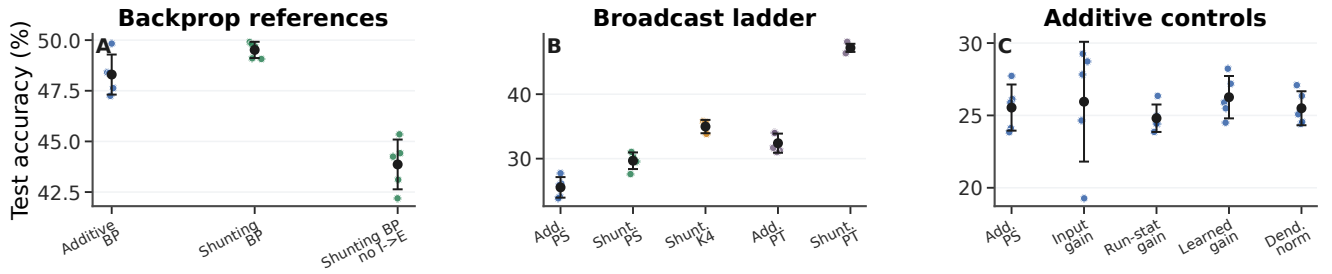


Figure S8: **CIFAR-10 control ladder in the compact direct-I-stream family.** (A) Matched backpropagation references. With nonnegative inputs and a nonlinear decoder, shunting slightly exceeds additive under standard training, and removing learned I-to-E conductance lowers the shunting reference. (B) Broadcast ladder. Per-soma feedback remains weak on this harder dataset, random low-rank  $K=4$  improves partially, and exact effective transported error nearly reaches the matched shunting backpropagation reference. (C) Additive fairness controls. Input-dependent gain, running-stat gain, learned gain, and dendritic normalization do not rescue additive per-soma feedback. This figure is a harder-dataset stress test, not a claim of competitive CIFAR-10 benchmarking.

| Condition                                | Core     | Training / broadcast | I-to-E | Test acc.                        |
|------------------------------------------|----------|----------------------|--------|----------------------------------|
| standard                                 | additive | backprop             | yes    | $48.3 \pm 1.0$                   |
| standard                                 | shunting | backprop             | yes    | <b><math>49.5 \pm 0.4</math></b> |
| standard, no learned inhibition          | shunting | backprop             | no     | $43.9 \pm 1.2$                   |
| per-soma                                 | additive | 5F per-soma          | yes    | $25.5 \pm 1.6$                   |
| per-soma, input-dependent gain           | additive | 5F per-soma          | yes    | $25.9 \pm 4.1$                   |
| per-soma, running-stat gain              | additive | 5F per-soma          | yes    | $24.8 \pm 0.9$                   |
| per-soma, learned gain                   | additive | 5F per-soma          | yes    | $26.3 \pm 1.5$                   |
| per-soma, dendritic normalization        | additive | 5F per-soma          | yes    | $25.5 \pm 1.2$                   |
| per-soma                                 | shunting | 5F per-soma          | yes    | $29.7 \pm 1.3$                   |
| per-soma, no learned inhibition          | shunting | 5F per-soma          | no     | $22.3 \pm 0.7$                   |
| transported error                        | additive | 5F path transport    | yes    | $32.4 \pm 1.5$                   |
| rank-4                                   | shunting | 5F low-rank          | yes    | $35.0 \pm 1.0$                   |
| transported error                        | shunting | 5F path transport    | yes    | <b><math>47.2 \pm 0.6</math></b> |
| transported error, no learned inhibition | shunting | 5F path transport    | no     | $38.6 \pm 1.0$                   |

Table S5: **Exact CIFAR-10 values for the 70-run control ladder (5 seeds per condition, validation-best epoch).** All rows use flattened CIFAR-10 in  $[0, 1]$ , nonnegative transfer outputs, positive conductance transforms, a compact depth-4 dendritic core, and a nonlinear decoder. LocalCA maps output error back to decoder-input/soma coordinates through the decoder Jacobian. The useful conclusions are narrow but important: learned shunting slightly improves the matched backpropagation reference, learned I-to-E conductance is necessary for the strongest shunting results, transported error nearly closes the shunting LocalCA gap, and additive per-soma fairness controls do not explain the transported-shunting result.

## A.12 Additive Normalization Control

This control separates generic additive voltage normalization from shunting conductance dynamics in a lower-capacity MNIST stress regime.

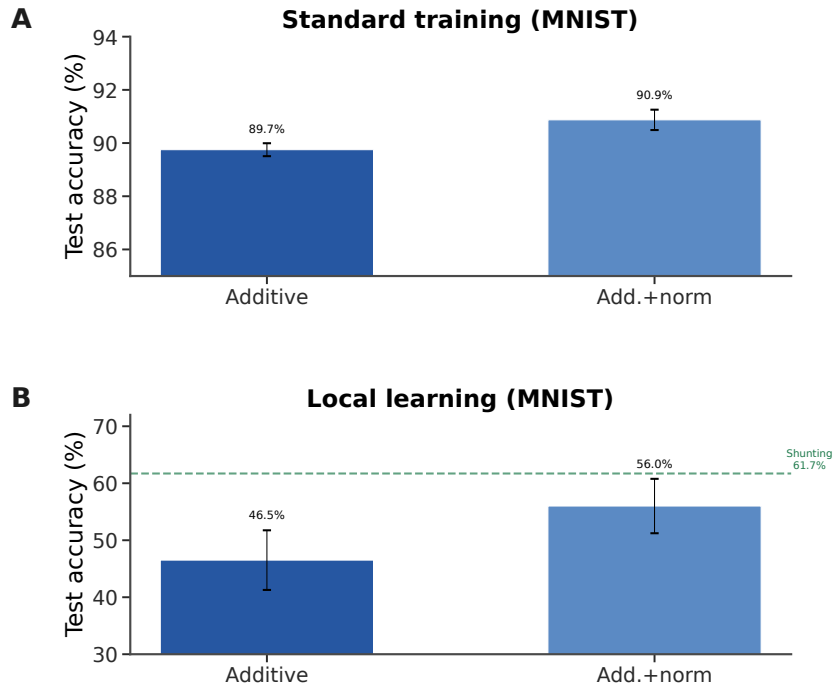


Figure S9: **Additive + normalization control (MNIST stress-regime diagnostic)**. **(A)** Standard backprop: normalization provides a small boost (89.7%  $\rightarrow$  90.9%). **(B)** Local learning: normalization improves additive from 46.5% to 56.0% (+9.5 pp), partially closing the gap to shunting (61.7%, dashed green). These lower local accuracies come from a separate stress-regime diagnostic, not from the matched-capacity performance setting in Table S4.

### A.13 Morphology $\times$ Inhibition Regime Map

This regime map (Fig. S10) tests whether tree geometry changes the inhibitory operating point at which shunting helps. It supports the claim that morphology changes the useful inhibition regime; a fuller mechanistic account would require matched theory diagnostics for every cell in the grid.

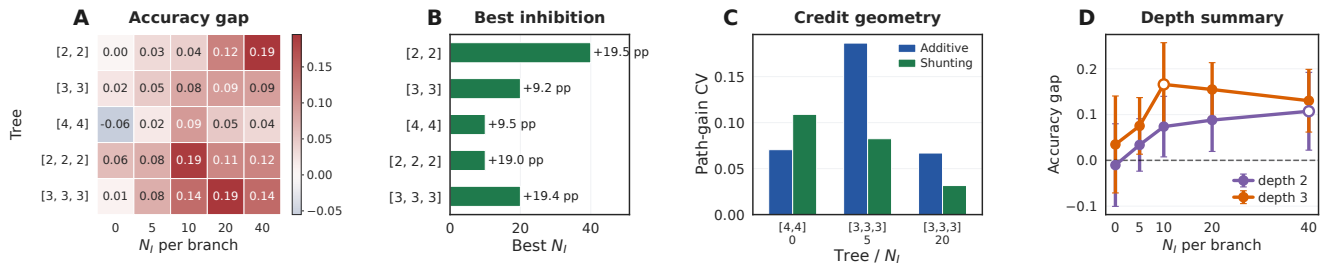


Figure S10: **Morphology shifts the useful inhibitory regime**. **(A)** Shunting-minus-additive LocalCA accuracy gap across dendritic morphologies and inhibitory synapse counts. **(B)** Best inhibitory count for each morphology. **(C)** Representative path-gain diagnostic for selected morphology cells, used to connect the performance map to credit geometry. **(D)** Average shunting advantage by depth, with markers showing the peak inhibitory count for each depth.

### A.14 Input-Mode Probe: Direct Inhibitory Stream vs. Explicit Inhibitory Cells

The main experiments use input-driven inhibitory streams: the transfer layer provides nonnegative excitatory and inhibitory channels from the external input, and excitatory dendrites receive learned I-to-E conductance banks directly. To test whether the same mechanism can be generated by an explicit feedforward inhibitory population, we ran a one-feedforward-layer noise-resilience probe that preserves the relevant dendritic depth ([3, 3] morphology). In this setting, explicit inhibitory cells reach  $94.2 \pm 0.2\%$  under path-transport LocalCA when their dendrites use the same

update policy as excitatory dendrites,  $93.8 \pm 0.2\%$  when those inhibitory-cell updates are frozen, and  $95.1 \pm 0.3\%$  under standard backprop. Direct input-driven inhibition gives the expected broadcast-fidelity ladder on the same one-layer architecture: per-soma feedback reaches  $85.2 \pm 0.7\%$ , while exact effective path transport reaches  $95.2 \pm 0.0\%$ . Thus explicit inhibitory cells can generate useful path-gain structure once the architecture isolates the dendritic-path question.

| Inhibitory input mode    | Training / broadcast                             | Test accuracy                       |
|--------------------------|--------------------------------------------------|-------------------------------------|
| Direct inhibitory stream | LocalCA, per-soma feedback                       | $0.852 \pm 0.007$                   |
| Direct inhibitory stream | LocalCA, exact effective path transport          | <b><math>0.952 \pm 0.000</math></b> |
| Explicit I cells         | LocalCA, path transport, train I-cell dendrites  | $0.942 \pm 0.002$                   |
| Explicit I cells         | LocalCA, path transport, freeze I-cell dendrites | $0.938 \pm 0.002$                   |
| Explicit I cells         | standard backprop                                | $0.951 \pm 0.003$                   |

Table S6: **One-feedforward-layer input-mode path-gain probe on noise resilience (3 seeds).** All rows use one excitatory dendritic population with  $[3, 3]$  morphology and nonnegative inputs. The direct-stream rows have no explicit inhibitory neurons; the input stream drives learned branch-level inhibitory conductance banks directly. The explicit-I rows use a matched inhibitory population with  $[3, 3]$  morphology. Exact effective path transport closes the direct-stream per-soma gap, and explicit inhibitory cells nearly match both direct-stream path transport and the explicit-I matched backpropagation reference.

**Post-training inhibition intervention and exact-error rank.** We next tested whether learned inhibitory conductance carries sample-dependent structure rather than only changing global scale (Fig. S11). We replayed trained shunting LocalCA checkpoints while replacing the branch-level inhibitory conductance with four post-training interventions: zero inhibition, sample-shuffled inhibition, per-branch batch means, or one uniform matched mean. On a 512-example held-out subset, learned MNIST inhibition at  $N_I=5$  gives  $90.4 \pm 1.1\%$  accuracy, while the four interventions drop to 19.8–26.4%. On noise resilience at  $N_I=10$ , learned inhibition gives  $80.5 \pm 2.4\%$ , while the interventions drop to 10.3–19.3%. Thus the inhibitory field is not interchangeable with a matched global conductance load. We also computed the SVD of exact compartment-error matrices across samples and compartments. At matched MNIST  $N_I=5$ , shunting has lower unrestricted rank-1 residual than additive ( $0.70 \pm 0.21$  vs.  $0.81 \pm 0.11$ ), lower participation rank ( $5.7 \pm 4.0$  vs.  $8.0 \pm 3.7$ ), and higher actual per-soma broadcast cosine ( $0.74 \pm 0.13$  vs.  $0.37 \pm 0.18$ ), supporting but not by itself proving the path-gain compressibility interpretation.

We also computed the direct log-gain covariance margin in Eq. (9) on the selected morphology diagnostic checkpoints. The margin is zero when no inhibitory conductance is present and slightly negative in the inhibited shunting selections ( $-2.5 \times 10^{-3}$ ,  $-5.0 \times 10^{-4}$ , and  $-2.8 \times 10^{-4}$  at  $N_I=5, 20, 40$ ). We therefore do not use this covariance condition as a positive main result. Its role is to clarify when shunting should compress gains; the empirical claim rests on the measured comparative path-gain dispersion, exact-error rank, and inhibition-intervention diagnostics.

| Model implication                                                                                                                                               | Biological prediction                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Path-local inhibitory conductance changes $\alpha_n^{\text{cond}}$ only for descendants whose credit path crosses the inhibited compartment.                    | Branch-local inhibitory perturbation should distort plasticity or teaching-signal efficacy for synapses below the perturbed compartment more than for sister branches. |
| The log derivative $\partial \log \alpha_n^{\text{cond}} / \partial G_k^I = -R_k^{\text{tot}}$ predicts stronger credit gating in high-resistance compartments. | The same inhibitory conductance change should have larger effects on low-conductance/high-resistance branches than on already high-conductance branches.               |
| Single-channel broadcast succeeds when exact compartment errors are effectively shared across the relevant branches and fails when branch identity matters.     | Tasks requiring route- or context-specific branch credit should need richer dendritic teaching signals than a single scalar modulatory broadcast.                      |

Table S7: **Experimental predictions from the path-gain mechanism.** These are qualitative biological predictions, not claims tested directly in this paper.

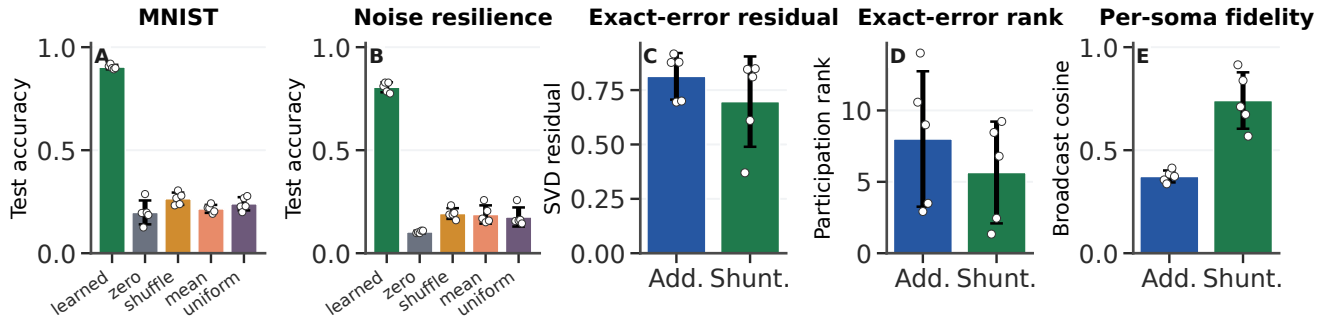


Figure S11: **Post-training inhibition intervention and exact-error compressibility.** (A,B) Post-training inhibitory-conductance interventions on trained shunting LocalCA checkpoints. Accuracy is measured on a 512-example held-out subset; dots show seeds. Removing, shuffling, or clamping learned inhibition collapses performance, showing that the learned inhibitory field is sample-dependent and functionally important. (C,D) Exact-error diagnostics on matched MNIST  $N_I=5$  runs. Shunting has a smaller unrestricted rank-1 residual, lower participation rank, and higher actual per-soma broadcast cosine than additive, so the exact error field is both more intrinsically compressible and better aligned with the implemented feedback field. Error bars are s.d. across 5 seeds.

## A.15 Experimental Predictions

The model is not a detailed cell-type circuit model, but its path-gain identity gives falsifiable dendritic predictions (Table S7). These predictions are naturally aligned with branch-level dendritic computation and compartment-targeted inhibition in pyramidal neurons [5, 6, 7, 13].

## A.16 Cue-Routing Feedback-Rank Diagnostic

Cue routing tests a regime where one shared teaching signal is expected to fail: context determines which noisy cue is reliable, so different pathways should receive different credit. Figure S12 shows that the benchmark is learnable and that moving beyond the restricted per-soma / single-channel baselines helps, but the path-structured variant does not yet beat the best unstructured low-rank setting. We therefore interpret this result as a failure-mode and rank-requirement diagnostic, not as a solved structured-feedback method.

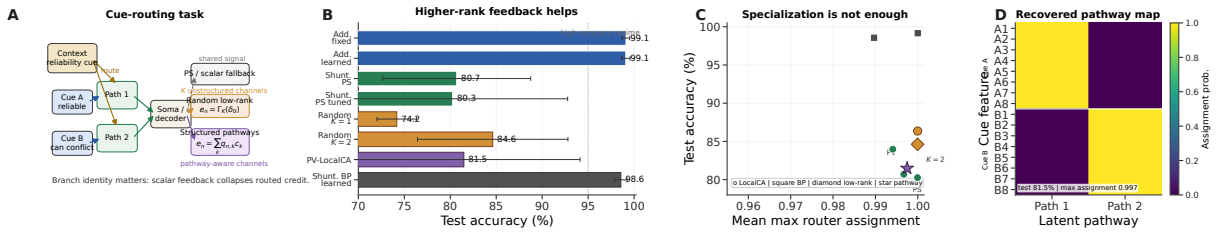


Figure S12: **Structured feedback when restricted feedback fails.** (A) Cue-routing task schematic. Context determines which noisy cue is reliable on each trial, so different dendritic pathways should receive different teaching signals. (B) Five-seed comparison of per-soma, random low-rank, and pathway-structured feedback. Moving beyond the per-soma / single-channel baselines helps, while the best structured repair remains open. (C) Test accuracy vs. router specialization across learned-router conditions. High specialization alone does not guarantee success. (D) Recovered pathway assignments for a representative learned-router run. Cue-A and cue-B features segregate into separate latent pathways even when feedback quality, rather than router discreteness, is the limiting factor.

## A.17 Random Low-Rank Broadcast Channels

These controls test whether failures of the restricted per-soma baseline reflect insufficient feedback bandwidth rather than a failure of local eligibility itself.

| Setting                                                           | Broadcast                      | Test accuracy                       |
|-------------------------------------------------------------------|--------------------------------|-------------------------------------|
| <i>Noise resilience, shunting, rank bridge</i>                    |                                |                                     |
| per-soma                                                          | per-soma feedback              | $0.662 \pm 0.016$                   |
| path propagation                                                  | recursive attenuation proxy    | $0.762 \pm 0.047$                   |
| low-rank                                                          | $K=1$                          | $0.453 \pm 0.126$                   |
| low-rank                                                          | $K=2$                          | $0.630 \pm 0.042$                   |
| low-rank                                                          | $K=4$                          | $0.764 \pm 0.025$                   |
| low-rank                                                          | $K=8$                          | $0.795 \pm 0.019$                   |
| path transport                                                    | exact effective path transport | <b><math>0.847 \pm 0.018</math></b> |
| <i>Cue routing, learned shunting router, rank/structure sweep</i> |                                |                                     |
| per-soma                                                          | tuned per-soma feedback        | $0.803 \pm 0.125$                   |
| low-rank                                                          | $K=1$                          | $0.742 \pm 0.022$                   |
| low-rank                                                          | $K=2$                          | <b><math>0.846 \pm 0.082</math></b> |
| low-rank                                                          | $K=4$                          | $0.748 \pm 0.085$                   |
| pathway-vector                                                    | tuned structured rank-2        | $0.815 \pm 0.126$                   |

Table S8: **Rank bridge and rank-structure comparison.** On shunting noise resilience, both recursive path propagation and higher-rank random low-rank broadcast close a substantial fraction of the gap between per-soma feedback and exact effective path transport, with low-rank continuing to improve up to  $K=8$ . On cue routing, the learned-router sweep shows that moving beyond the per-soma baseline helps, but the best unstructured low-rank setting ( $K=2$ ) is still slightly better than the tuned structured pathway-vector variant. We therefore interpret routed credit assignment as an open higher-rank problem rather than as a solved pathway-vector rescue.

## B Dendritic Network Implementation

This section gives the implementation-level specification needed to reproduce the dendritic architectures and LocalCA updates. It separates the biological modeling assumptions from software choices such as positive raw-parameter transforms, decoder-error coordinates, and optimizer gradient assignment.

## B.1 Implementation Overview

Each feedforward dendritic layer is a population of dendritic neurons. A neuron contains a rooted tree of compartments; each compartment has an excitatory synapse bank, optionally an inhibitory synapse bank, and dendritic coupling conductances from its children. The main experiments use branch-level input-driven inhibition: a nonnegative transfer stage produces excitatory and inhibitory input streams from the external input, and the inhibitory stream drives learned I-to-E conductance banks on excitatory dendritic branches. The explicit-I probe in Table S6 instead instantiates a separate feedforward inhibitory population, whose output projects inhibitory conductance onto excitatory dendrites.

The additive and shunting cores are architecture-matched: they use the same tree, synapse counts, nonnegative transfer stage, positive conductance parameterization, optimizer protocol, and seed sets. They differ only in how a branch combines excitatory, inhibitory, and dendritic inputs. Decoder weights are unconstrained readout parameters and are not interpreted as conductances. Table S9 maps each implementation concept to its governing equation, code location, and update scope.

| Concept / parameter family                       | Paper eq.                                                                  | Code location                                                                                                                                                                                              | Update scope / notes                                                                                      |
|--------------------------------------------------|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Shunting branch voltage                          | Eq. (1)                                                                    | <code>DendriticBranchLayer.forward</code><br>( <code>use_shunting=True</code> )                                                                                                                            | Numerator/denominator conductance integration; in-theorem.                                                |
| Additive control voltage                         | Eq. (1)                                                                    | <code>DendriticBranchLayer.forward</code><br>( <code>use_shunting=False</code> );<br><code>normalize_additive_voltage</code>                                                                               | Same tree/synapses; inhibition subtractive rather than a denominator load.                                |
| E/I synaptic and dendritic-coupling conductances | Cor. 1;<br>Eqs. (13)–(14)                                                  | <code>LocalCreditAssignment._apply_local_rule_gradients</code> (conductance branch)                                                                                                                        | LocalCA eligibility $xR^{\text{tot}}(E-V)e$ , then positive-transform derivative; covered by the theorem. |
| Additive eligibility (matched control)           | Eq. (13)                                                                   | <code>_apply_local_rule_gradients</code> (additive branch)                                                                                                                                                 | Matched signed additive derivative, with optional additive-gain controls.                                 |
| Broadcast-mode switch                            | broadcast $e_n$                                                            | <code>LocalRuleConfig.error_broadcast_mode</code> ;<br><code>_compute_low_rank_broadcast</code> ;<br><code>_precompute_path_transport_errors</code> ;<br><code>_precompute_path_propagation_factors</code> | Selects scalar/per-soma, low-rank, path-transport, or pathway-vector feedback; bandwidth-ladder controls. |
| 4F / 5F reliability gates                        | $r_n^{4F}, \phi_n$                                                         | <code>_compute_layer_rho</code> , <code>_compute_layer_phi*</code> ,<br><code>clamp_phi</code> , <code>FiveFactorConfig</code>                                                                             | Slow bounded preconditioners, not additional task-error channels; outside the theorem.                    |
| Reactivation transfer                            | $f_n(V)$                                                                   | configured local/autograd schedule                                                                                                                                                                         | Post-voltage activation derivative; implementation-level training component.                              |
| Top- $k$ synapse banks                           | —                                                                          | <code>masked_apply_local_rule_gradients</code>                                                                                                                                                             | Sparse input selection; masks multiply the branch eligibility.                                            |
| Linear / nonlinear decoder                       | $\delta_0^a = W_{\text{dec}}^\top \delta^y / J_{\text{dec}}^\top \delta^y$ | local decoder gradient or autograd                                                                                                                                                                         | Maps output error into soma/core coordinates; implementation-level.                                       |
| HSIC auxiliary                                   | —                                                                          | auxiliary gradient on selected representations                                                                                                                                                             | Batch dependence regularizer; auxiliary objective, not a local theorem term.                              |
| Router / pathway roles                           | pathway-vector                                                             | configuration-dependent router/pathway updates                                                                                                                                                             | Exploratory higher-rank routing; outside the theorem.                                                     |
| Experiment knobs                                 | —                                                                          | <code>LocalRuleConfig</code> YAML ( <code>rule_variant</code> ,<br><code>error_broadcast_mode</code> , <code>broadcast_rank</code> ,<br><code>five_factor</code> )                                         | Reproduces rule family, feedback bandwidth, and 5F stabilizer settings.                                   |

Table S9: **Implementation map: concepts, equations, code, and update scope.** The exact factorization (Cor. 1) covers conductance-stage branch and dendritic-coupling parameters; decoder, reactivation, top- $k$ , HSIC, and router mechanisms are implementation-level training components documented alongside the theorem rather than derived from it. Code anchors are symbolic names so that minor refactors do not invalidate the map.

**Implementation-specific update paths.** Top- $k$  synapse-bank scores are learned selection parameters; the selected masks gate active presynaptic drives and the corresponding local gradients, while inactive weights receive no LocalCA update unless an explicit inactive-update control is enabled. The discrete selection itself is treated as an implementation-level sparsity mechanism rather than as a conductance variable in the theorem. Reactivation slope and bias parameters use the configured optimizer schedule; the conductance update still uses the local derivative  $f'_n(V_n)$  for voltage-error conversion. Router/pathway parameters in the cue-routing controls are trained through their configured router/pathway objective and are not included in the conductance-factorization theorem. HSIC gradients

are applied to selected representation tensors through the configured local-learning representation pathway; they do not replace the LocalCA conductance eligibility and are reported as an auxiliary objective.

## B.2 Morphology and Synapse Banks

The morphology notation  $[b_1, b_2, \dots, b_D]$  specifies a rooted dendritic tree. The factor  $b_d$  is the fan-in from dendritic stage  $d$  to the next more proximal stage. Thus  $[3, 3]$  gives 3 proximal branches per soma and 3 distal branches per proximal branch, for 9 distal leaves per soma. In the main feedforward models, synaptic inputs terminate on dendritic branches rather than directly on the soma. Unless a control explicitly changes this, each branch has  $N_E$  excitatory synapses and  $N_I$  inhibitory synapses. The branch input is sparse through a learned top- $k$  or banked synapse selection module; inactive synapses are masked out in the local update unless the corresponding control enables inactive-weight updates.

For a branch  $n$ , write  $x_{nj}^E, x_{nj}^I \geq 0$  for the selected excitatory and inhibitory presynaptic drives and  $a_c$  for the post-reactivation activity of child compartment  $c$ . The branch-level excitatory, inhibitory, and dendritic conductance currents are

$$U_n^E = \sum_{j=1}^{N_E} g_{nj}^E x_{nj}^E, \quad U_n^I = \sum_{j=1}^{N_I} g_{nj}^I x_{nj}^I, \quad D_n = \sum_{c \in \text{child}(n)} g_{c \rightarrow n}^{\text{den}} a_c.$$

The corresponding dendritic coupling load is

$$G_n^{\text{den}} = \sum_{c \in \text{child}(n)} g_{c \rightarrow n}^{\text{den}}.$$

For leaf branches,  $D_n = G_n^{\text{den}} = 0$ . For direct input-driven inhibition,  $x^I$  is produced by the inhibitory transfer stream. For explicit-I controls,  $x^I$  is the activity of the inhibitory-cell population.

## B.3 Shunting and Additive Forward Equations

In the shunting core, excitatory reversal is 1, leak and inhibitory reversal are 0, and leak conductance is fixed to 1. The implemented branch voltage is therefore

$$V_n^{\text{shunt}} = \frac{U_n^E + D_n}{1 + U_n^E + U_n^I + G_n^{\text{den}} + \varepsilon}.$$

The small numerical  $\varepsilon$  is used only for floating-point safety; analytically the leak conductance keeps the denominator positive. In implementation equations, the same stabilized denominator is used for the recorded input resistance:

$$g_n^{\text{tot}} = 1 + U_n^E + U_n^I + G_n^{\text{den}}, \quad g_{n,\varepsilon}^{\text{tot}} = g_n^{\text{tot}} + \varepsilon, \quad R_{n,\varepsilon}^{\text{tot}} = (g_{n,\varepsilon}^{\text{tot}})^{-1}.$$

The main analytical equations write  $R_n^{\text{tot}} = (g_n^{\text{tot}})^{-1}$ ; exact-gradient reconstruction tests use the implementation convention above. In the additive control, inhibition is subtractive rather than shunting. With the same nonnegative conductance variables and nonnegative inputs, the implemented comparator is

$$V_n^{\text{add}} = U_n^E + D_n - U_n^I,$$

with optional additive-normalization controls reported separately in Fig. S9. Thus additive inhibition changes the signed voltage contribution and local additive eligibility, but it does not enter the denominator and does not gate conductance-stage path gain through  $R_n^{\text{tot}}$ ; Table S10 summarizes the matched comparison.

| Property              | Shunting / conductance core                               | Additive control                                      |
|-----------------------|-----------------------------------------------------------|-------------------------------------------------------|
| Voltage               | Normalized by total conductance                           | Signed E/I voltage sum                                |
| Inhibition            | Divisive conductance load                                 | Subtractive signed term                               |
| Synaptic eligibility  | $xR^{\text{tot}}(E - V)e$                                 | $sxe, s \in \{+1, -1\}$                               |
| Driving force         | Present through $E - V$                                   | Absent from additive derivative                       |
| Input resistance      | State-dependent, $R^{\text{tot}} = (g^{\text{tot}})^{-1}$ | Effectively fixed                                     |
| Credit interpretation | Inhibition changes path gain and voltage sensitivity      | Inhibition shifts voltage but does not gate path gain |

Table S10: **Matched shunting/additive comparison.** Both architectures use their own local derivative, so the comparison changes the forward-pass integration rule rather than applying a shunting-derived update to an additive model.

## B.4 Positive Parameterization and Raw-Parameter Gradients

Trainable synaptic and dendritic conductances are stored as unconstrained raw parameters  $\theta$  and transformed by a positive map  $g = \psi(\theta)$ , with `softplus` used in the main reported runs:

$$\psi(\theta) = \log(1 + \exp \theta), \quad \psi'(\theta) = \sigma(\theta).$$

Equations in the main text give conductance-space gradients  $\partial L / \partial g$ . Before assigning gradients to the optimizer variables, the implementation applies the chain rule

$$\frac{\partial L}{\partial \theta} = \frac{\partial L}{\partial g} \psi'(\theta).$$

The exact-gradient reconstruction diagnostic includes this parameterization factor, as well as top- $k$  activity masks and dendritic block-linear parameterization. This is why the reconstruction comparison is against raw autograd gradients rather than only against abstract conductance variables.

## B.5 Post-Voltage Reactivation

The conductance stage produces a pre-reactivation voltage  $V_n$ , after which each branch may transmit a monotone post-voltage activity  $a_n = f_n(V_n)$ . The main sweeps use the learnable bounded transform

$$f_n(V) = \frac{\tanh(m_n(V - b_n)) + 1}{2}, \quad m_n = \exp(\ell_{m,n}),$$

with derivative

$$f'_n(V) = \frac{m_n}{2} [1 - \tanh^2(m_n(V - b_n))].$$

When reactivation is enabled, exact path transport uses the effective gain  $\tilde{\alpha}_n$ : parent-to-child transport includes the parent activation derivative, and the local update converts activation-space error into pre-reactivation voltage error by multiplying by the child derivative. With identity transfer, this reduces to conductance-stage transport through  $\alpha_n^{\text{cond}}$ .

## B.6 Decoder and Error Coordinates

Let  $h$  denote the final soma/core activity passed to the decoder and let  $\delta^y = \partial L / \partial \hat{y}$ . For a linear decoder  $\hat{y} = W_{\text{dec}} h$ , the soma/core activation-space teaching error is

$$\delta_0^a = W_{\text{dec}}^\top \delta^y.$$

For a nonlinear decoder, used in the CIFAR-10 compact control ladder, the implementation uses the decoder-input Jacobian product

$$\delta_0^a = J_{\text{dec}}(h)^\top \delta^y.$$

| Quantity         | Symbol                           | Convention                                                                                                             |
|------------------|----------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Voltage          | $V$                              | Conductance-stage shunting voltage in $[0, 1]$ under nonnegative drive; additive controls may be signed                |
| Conductances     | $g^{\text{syn}}, g^{\text{den}}$ | Nonnegative via $g = \text{softplus}(\theta)$ ; local gradients are mapped to raw $\theta$ by the transform derivative |
| Leak conductance | $g^{\text{leak}}$                | Set to 1                                                                                                               |
| Input resistance | $R^{\text{tot}}$                 | $\leq 1$                                                                                                               |

Table S11: Units and normalization.

All LocalCA broadcast modes are defined in this soma/core activation-error coordinate system before local conversion to voltage-space errors. Decoder-update modes are: **backprop**, which uses autograd for decoder parameters; **local**, which is implemented for the linear readout and assigns

$$\widehat{\mathbf{V}}_{W_{\text{dec}}} L = \left\langle \delta^y h^\top \right\rangle_B, \quad \widehat{\mathbf{V}}_{b_{\text{dec}}} L = \langle \delta^y \rangle_B;$$

and **frozen**, which holds the decoder fixed. The local decoder gradient is output-space;  $\delta_0^a$  is the mapped soma/core activation-space error used by dendritic broadcasts.

## B.7 Biological Plausibility Assumptions

The model makes the following explicit assumptions. First, it uses the steady-state solution of the passive cable equation rather than temporal membrane dynamics; all path gains and gradient diagnostics refer to conductance-stage voltages at steady state. Second, each dendritic cell is a rooted tree, so each compartment has a unique path to the soma and the closed-form path gains  $\alpha_n^{\text{cond}}$  and  $\tilde{\alpha}_n$  are well-defined. Third, synaptic and dendritic conductance parameters are nonnegative by construction. Fourth, the main sweeps enforce nonnegative first-layer presynaptic drive through nonnegative inputs or a ReLU transfer stage. Fifth, post-voltage reactivation is handled through local  $f'(V)$  factors along the path. Sixth, each dendritic synapse receives a restricted non-local feedback field–scalar, per-soma, low-rank, path-structured, or transported-oracle depending on the condition—while presynaptic activity, voltage, reversal potential, input resistance, and slowly estimated branch-level modulators are local or branch-local quantities.

## B.8 Units and Parameterization

Table S11 summarizes the numerical conventions used throughout the implementation and diagnostics, including which quantities are constrained conductances and which are unconstrained readout parameters.

## B.9 Representative Manuscript Architectures

| Setting                                                    | Encoder  | E layers           | Tree                                               | Synapses / branch                                             | Seeds                   | Notes                                                                                                                                              |
|------------------------------------------------------------|----------|--------------------|----------------------------------------------------|---------------------------------------------------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Gradient fidelity / path-gain / noise-resilience mechanism | id.+ReLU | [128, 128]         | [3, 3]                                             | $N_E=40$ ,<br>$N_I \in \{0, 5, 10, 20, 40\}$                  | 5 fidelity,<br>5 oracle | Main mechanism sweeps in Figs. 2 and 3; reactivation <code>param_tanh</code> .                                                                     |
| MNIST / figure-ground local performance                    | id.+ReLU | [128]              | [3, 3]                                             | $N_E=40$ , $N_I=20$                                           | 5                       | Best local 5F per-soma runs in Table S4; reactivation <code>param_tanh</code> .                                                                    |
| Fashion-MNIST performance                                  | id.+ReLU | [128]              | [3, 3]                                             | $N_E=40$ , $N_I=20$                                           | 5                       | Dedicated five-seed standard-vs-local performance sweep; reactivation <code>param_tanh</code> .                                                    |
| Cue-routing PV-LocalCA                                     | router   | [64]               | [2]                                                | $N_E=10$ , $N_I=4$                                            | 5                       | Learned router with two pathway groups and rank-2 structured feedback; reactivation <code>param_tanh</code> .                                      |
| CIFAR-10 compact direct-I-stream control ladder            | id.+ReLU | [20] E, no I cells | [3, 3, 3, 3]                                       | $N_E=25$ , $N_I=25$ direct I-to-E, with matched no-I controls | 5 per condition         | Harder-dataset stress test with input-driven inhibitory conductance, decoder [32, 16], decoder-aware soma mapping, and additive fairness controls. |
| Morphology $\times$ inhibition appendix map                | id.+ReLU | [128, 128]         | [2, 2], [3, 3],<br>[4, 4], [2, 2, 2],<br>[3, 3, 3] | $N_E=40$ ,<br>$N_I \in \{0, 5, 10, 20, 40\}$                  | 3                       | Supportive 3-seed sweep; reactivation <code>param_tanh</code> .                                                                                    |

Table S12: **Representative architectures used in the manuscript.** All dendritic rows use explicit branch-level excitatory and inhibitory synapse banks without direct somatic synapses, nonnegative conductances, and a nonnegative first-layer drive enforced by an explicit ReLU transfer stage in the main runs. The tree column gives dendritic morphology per excitatory neuron, and the synapse counts report excitatory and inhibitory synapses per branch.

## B.10 Hyperparameters

Table S13 collects the core training hyperparameters for each main experiment. Learning rates are applied through parameter groups (top- $k$ , dendritic block-linear, reactivation, decoder) with the values given in the representative config files; where a single LR is reported, every group uses that value.

| Setting                                                                  | Opt. | LR                                             | Batch | Epochs | Weight decay | Notes                                                                |
|--------------------------------------------------------------------------|------|------------------------------------------------|-------|--------|--------------|----------------------------------------------------------------------|
| MNIST / Fashion-MNIST / figure-ground MNIST (5F per-soma LocalCA)        | Adam | $10^{-3}$ / $5 \cdot 10^{-4}$ (block / react.) | 256   | 100    | 0            | fixed-epoch, no early stopping; <code>param_tanh</code> reactivation |
| MNIST / Fashion-MNIST / figure-ground MNIST (matched backprop reference) | Adam | $10^{-3}$ / $5 \cdot 10^{-4}$                  | 256   | 100    | 0            | matched to LocalCA setup                                             |
| Gradient fidelity / path-gain / noise resilience                         | Adam | $10^{-3}$                                      | 256   | 50     | 0            | 5 seeds, hooks enabled for diagnostic capture                        |
| Morphology $\times$ inhibition regime map (supportive)                   | Adam | $10^{-3}$                                      | 256   | 50     | 0            | 3 seeds                                                              |
| Cue routing                                                              | Adam | $10^{-3}$                                      | 256   | 90     | 0            | early stopping with patience 30                                      |
| CIFAR-10 compact direct-I-stream depth-4 LocalCA                         | Adam | $10^{-3}$ / $10^{-4}$ (block / react.)         | 256   | 400    | 0 / 0.01     | grad-clip 5.0, early stop patience 50, nonlinear decoder             |
| CIFAR-10 compact direct-I-stream depth-4 BP                              | Adam | $10^{-3}$                                      | 256   | 200    | 0.01         | early stop patience 40, matched backpropagation reference            |

Table S13: **Training hyperparameters by experiment.** All runs use Adam with default  $\beta_1=0.9$ ,  $\beta_2=0.999$ ,  $\epsilon=10^{-8}$ . Reactivation and block-linear parameter groups use a slower LR to keep conductance-level dynamics stable. Representative configs are provided in the accompanying repository.

## B.11 Effective Broadcast Dimensionality in Main Architectures

Table S14 reports the effective feedback dimensionality of each main architecture.

| Setting                                          | Soma/core dim. | Output classes | Main feedback field                                                                                           |
|--------------------------------------------------|----------------|----------------|---------------------------------------------------------------------------------------------------------------|
| Gradient fidelity / path-gain / noise resilience | 128, 128       | 10             | Per-soma soma/core error at matched-width dendritic layers                                                    |
| MNIST / Fashion-MNIST performance                | 128            | 10             | Per-soma soma/core error                                                                                      |
| Figure-ground MNIST performance                  | 128            | 10             | Per-soma soma/core error                                                                                      |
| Cue routing                                      | 64             | 2              | Tuned per-soma baseline; higher-rank tests use low-rank or pathway-vector feedback                            |
| Compact CIFAR-10 harder-dataset stress test      | 20             | 10             | Vector-valued per-soma after decoder-aware mapping; higher-rank tests use explicit low-rank or path transport |

Table S14: **Effective dimensionality of the main feedback in the reported architectures.** Output loss gradients are first mapped into decoder-input/soma coordinates. The main performance and mechanism runs use the implemented `per_soma` mode when the soma/core dimension matches layer width; scalar, low-rank, pathway-vector, and transported-error conditions are labeled separately as controls or oracles.

## B.12 Compute Resources

All experiments were run on an institutional GPU cluster. Reproduction runs use one NVIDIA A100 or H100-class GPU (40–80GB memory) per seed and do not require distributed training. Table S15 gives approximate single-GPU runtime ranges for the reported experiment families; unreported exploratory runs used additional cluster time and are not counted in the reproduction estimate.

| Experiment family                                       | Runs | Time / run  | Approx. GPU-h |
|---------------------------------------------------------|------|-------------|---------------|
| Performance and verification                            | ~25  | 5–20 min    | 4–9           |
| Gradient, path-gain, inhibition, and oracle diagnostics | ~80  | 10–30 min   | 15–40         |
| Morphology, stress, rule, and feedback controls         | ~120 | 5–35 min    | 20–70         |
| CIFAR-10 control ladder                                 | 70   | 0.8–2.5 h   | 60–175        |
| Figure generation and CPU-side summaries                | –    | < 5 h total | < 5           |

Table S15: **Approximate compute for reported results.** Ranges reflect single-seed wall-clock variation across A100/H100-class GPUs, data-loading overhead, and early stopping.

## B.13 Code, Data, and Asset Availability

MNIST, Fashion-MNIST, and CIFAR-10 are public datasets or public benchmark assets cited in the paper; we use them under their standard published access conditions and cite their original sources in the bibliography. The accompanying source package contains the core model, training code, diagnostic utilities, representative configuration files, tests, figure-source summaries, and analysis entry points. Code will be released publicly at the time of arXiv posting with an immutable tag. The paper does not release a new dataset or a stand-alone pre-trained model asset. Dataset access pages are the original MNIST site / UCI entry (<https://archive.ics.uci.edu/dataset/683/mnist+database+of+handwritten+digits>), the Fashion-MNIST repository (<https://fashion-mnist.s3.amazonaws.com/>).

[//github.com/zalando-research/fashion-mnist](https://github.com/zalando-research/fashion-mnist)), and the CIFAR page (<https://www.cs.toronto.edu/~kriz/cifar.html>).

| Asset           | Use in this paper                                            | Access / license or terms note                                                                                                                                                     |
|-----------------|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MNIST           | Digit classification and derived nonnegative synthetic tasks | Public benchmark from the original MNIST site / UCI entry; UCI lists DOI 10.24432/C53K8Q and asks users to follow the original acknowledgement policy; no raw-data redistribution. |
| Fashion-MNIST   | Apparel classification stress test                           | Public Zalando Research benchmark; repository is MIT licensed; no raw-data redistribution beyond standard dataset loaders.                                                         |
| CIFAR-10        | Flattened harder-dataset stress test                         | Public CIFAR dataset site asks users to cite Krizhevsky’s technical report; no raw-data redistribution.                                                                            |
| Synthetic tasks | Figure-ground MNIST, noise resilience, cue integration       | Generated procedurally from public benchmarks or random seeds described in Appendix D; no new third-party asset is introduced.                                                     |

Table S16: **Existing assets used in the manuscript.** We credit the original sources in the bibliography, use public benchmark assets under their published access conditions, and do not redistribute third-party raw data in the manuscript package.

## B.14 LocalCA Broadcast and Gradient Assignment

LocalCA first maps the output loss derivative to the soma/core activation-space error  $\delta_0^a$  and then constructs a branch voltage-error field. In the final dendritic layer this is the decoder-input error; in earlier matched-width layers the practical implementation reuses the same coordinate by neuron index as an approximate layer-soma teaching signal. This is an inter-layer feedback approximation, separate from the within-tree path-transport approximation studied by the main theory. In the main reported `per_soma` mode, if the layer width matches the soma/core error dimension, each soma-specific error component is broadcast to the corresponding dendritic compartments:

$$e_{n,b}^a = \delta_{0,b}^a \in \mathbb{R}^{d_0}.$$

The stricter scalar control compresses the soma/core error to one value per example,

$$\bar{\delta}_b = \frac{1}{d_0} \sum_{c=1}^{d_0} \delta_{0,bc}^a, \quad e_{n,b}^a = \bar{\delta}_b \mathbf{1}_n,$$

where  $d_0$  is the dimension of the soma/core error. If dimensions do not match, the implementation falls back to scalar-style expansion rather than silently inventing an unmatched per-soma map. The low-rank controls use fixed random projection and mixing matrices  $P_K$  and  $Q_n$ ,

$$c_b = P_K \delta_{0,b}^a \in \mathbb{R}^K, \quad e_{n,b}^a = Q_n c_b.$$

The pathway-vector controls use router-inferred pathway roles to gate and transport a low-rank role vector. The transported oracle uses the effective tree recursion. If  $p$  is the parent of child  $c$ , activation-space error is propagated by

$$e_c^a = e_p^a f'_p(V_p) R_{p,\varepsilon}^{\text{tot}} g_{c \rightarrow p}^{\text{den}},$$

and the local pre-reactivation voltage error used in conductance eligibility is

$$e_n^V = \begin{cases} e_n^a f'_n(V_n), & \text{post-voltage reactivation enabled,} \\ e_n^a, & \text{identity transfer.} \end{cases}$$

This operational definition is the implementation counterpart of the effective path gain  $\tilde{\alpha}_n$  in Theorem 1.

For shunting branches, the conductance-space LocalCA gradients assigned before raw-parameter transformation are

$$\hat{\nabla}_{g_{nj}^E} L = \langle e_n^V x_{nj}^E R_{n,\varepsilon}^{\text{tot}} (1 - V_n) \rangle_B, \quad \hat{\nabla}_{g_{nj}^I} L = \langle e_n^V x_{nj}^I R_{n,\varepsilon}^{\text{tot}} (0 - V_n) \rangle_B,$$

and for a dendritic coupling from child  $c$  to parent  $n$ ,

$$\hat{\nabla}_{g_{c \rightarrow n}^{\text{den}}} L = \langle e_n^V R_{n,\varepsilon}^{\text{tot}} (a_c - V_n) \rangle_B.$$

For additive controls, the matching local derivatives are

$$\hat{\nabla}_{g_{nj}^E} L = \langle e_n^V x_{nj}^E \rangle_B, \quad \hat{\nabla}_{g_{nj}^I} L = -\langle e_n^V x_{nj}^I \rangle_B, \quad \hat{\nabla}_{g_{c \rightarrow n}^{\text{den}}} L = \langle e_n^V a_c \rangle_B.$$

The optimizer applies the usual descent step; equivalently, the sign can be absorbed into the definition of the broadcast error. After these conductance-space gradients are formed, they are multiplied by the positive-transform derivative  $\psi'(\theta)$  and by any active synapse masks before assignment to raw parameters.

### B.15 4F and 5F Local Modulators

The 3F rule uses only the local eligibility and broadcast error above. The 4F and 5F variants multiply the conductance-space gradient by branch-level reliability factors before raw-parameter transformation. For 4F, the implementation records  $V_n$  from each branch layer and the corresponding soma/core activity  $V_0$ . For each batch item  $b$ , it first averages over the non-batch coordinates of the recorded layer tensor to obtain scalar summaries  $\bar{V}_{n,b}$  and  $\bar{V}_{0,b}$ . It then estimates covariance over the batch axis and smooths the resulting scalar with an EMA:

$$r_n^{4F} = \frac{\text{Cov}_B(\bar{V}_{n,b}, \bar{V}_{0,b})}{\sqrt{\text{Var}_B(\bar{V}_{n,b}) \text{Var}_B(\bar{V}_{0,b}) + \varepsilon}}.$$

The EMA is initialized from the first observed batch statistic; the recorded tensors are detached from autograd, and the same mini-batch supplies both the statistics and the gradient update. This proxy is clamped to the positive stability range  $[0.1, 2.0]$  before it multiplies the local gradient; sign-opposed covariance therefore reduces the multiplier rather than reversing update direction. Because the smoothed numerator and denominator can be updated separately, and because single-sample fallbacks use online covariance moments, the clamped value can exceed one and should be read as a preconditioner rather than a literal Pearson correlation. Single-sample online fallbacks for these statistics use Welford’s algorithm [25]. The 5F factor adds a bounded branch-level preconditioner based on how much branch voltage variance remains predictable from parent or soma-level activity:

$$\phi_n = \text{clip}_{[0.25, 4.0]} \left( \frac{\text{Var}(V_n)}{\sigma_{\text{res},n}^2 + \varepsilon} \right),$$

where  $\sigma_{\text{res},n}^2$  is estimated online from a scalar ridge regression of the branch-layer voltage on the configured parent proxy. Batch-mode estimates center over examples and flatten remaining layer coordinates before updating the smoothed variance and covariance moments; single-sample fallback uses scalar online moments. Since these moments are smoothed and clamped rather than recomputed as a single ordinary least-squares fit on a fixed batch,  $\phi_n$  can transiently fall below one or rise above one. The lower clamp keeps noisy residual estimates from suppressing a branch-layer update entirely, and the upper clamp prevents highly predictable branches from dominating optimization. In the main 5F runs, the multiplier is  $r_n^{4F} \phi_n$ . The 5F sensitivity diagnostic in Fig. S2 varies the clamp and EMA rate to verify that the reported MNIST performance is not a single brittle clamp setting. By default the reliability statistics use EMA rate  $\alpha_{\text{EMA}} = 0.1$  and a conditional-EMA residual estimator with ridge  $\lambda_{\text{ridge}} = 10^{-3}$  for  $\sigma_{\text{res},n}^2$ , and all gate denominators use the stabilizer  $\varepsilon = 10^{-8}$ .

## B.16 Algorithm

Algorithm 1 gives the update order for the main 5F per-soma LocalCA condition and shows where scalar, higher-bandwidth, and transported-error controls enter.

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### Algorithm 1 Main 5F Per-Soma LocalCA Update

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- 1: **Input:** Dendritic model, batch  $(x, y)$ , optimizer
  - 2: Forward pass; loss  $L$ , output error  $\delta^y$
  - 3: Somatic/core activation error  $\delta_0^a = \text{decoder-input Jacobian product } J_{\text{dec}}(h_{\text{core}})^\top \delta^y$  (or  $W_{\text{dec}}^\top \delta^y$  for a linear decoder)
  - 4: **for** each layer  $n$  (reverse) **do**
  - 5:   Per-soma feedback  $e_n^a = \delta_0^a$  when dimensions match; scalar/low-rank/path controls replace this line
  - 6:   Update EMA estimates for  $r_n^{4F}$  and  $\phi_n$  from batch voltages
  - 7:   Convert to pre-reactivation voltage error  $e_n^V$  using the local activation derivative when needed
  - 8:    $\widehat{V}_{g_j^{\text{syn}}} L \leftarrow r_n^{4F} \phi_n \langle x_j R_{n,\epsilon}^{\text{tot}} (E_j - V_n) e_n^V \rangle_B$
  - 9:    $\widehat{V}_{g_{c \rightarrow n}^{\text{den}}} L \leftarrow r_n^{4F} \phi_n \langle R_{n,\epsilon}^{\text{tot}} (a_c - V_n) e_n^V \rangle_B$
  - 10:   Map conductance gradients to raw parameters by multiplying by  $\partial g / \partial \theta$
  - 11: **end for**
  - 12: Optional controls replace line 5 with scalar, rank- $K$ , path-structured, or transported-oracle feedback.
  - 13: Clip gradients; optimizer step
- 

## C Theoretical Details

This section collects theory-adjacent material that supports implementation and interpretation but is not part of the main proof. The central derivation remains Theorem 1, Corollary 1, and Prop. 2.

### C.1 Random-Broadcast Alignment Intuition

Feedback-alignment arguments suggest that random or low-rank feedback can be useful when the induced local update remains positively correlated with the exact gradient [14]. In this model, that correlation depends on how local eligibility factors co-vary with the conductance-stage path gain in Eq. (4). We treat this as intuition only. The central theoretical object in the paper is still the exact pre-reactivation compartment error  $\partial L / \partial V_n$ ; under identity upward transfer this reduces to  $\alpha_n^{\text{cond}} \delta_0^V$ , while post-voltage activation uses  $\tilde{\alpha}_n \delta_0^V$ . The empirical question is how faithfully different broadcast fields approximate that quantity.

### C.2 Morphology-Aware Extensions

The following variants are implemented controls or architectural extensions used to test whether coarse morphology-aware feedback can approximate exact path transport. They are not required for the main 5F per-soma results, but they define the path-propagation, depth, normalization, and pathway-vector controls reported in the appendix figures and tables.

**Path-integrated propagation.** Modulate broadcast error by  $\pi_n = \pi_{n-1} \cdot R_{n-1}^{\text{tot}} \cdot \bar{g}_{n-1}^{\text{den}}$ , approximating depth attenuation from Eq. (4) without computing the exact sample-specific path gain.

**Depth modulation.** Per-branch scaling  $\kappa_j = \kappa_{\text{base}} / (d_j + d_0)$ , mirroring cable attenuation and testing whether a simple depth prior can stabilize distal updates.

**Dendritic normalization.**  $\Delta g_j^{\text{den}} \leftarrow \Delta g_j^{\text{den}} / (\sum_k g_k^{\text{den}} + \epsilon)$ , analogous to homeostatic scaling [26] and used as an additive-control normalization comparison.

**Pathway-vector feedback.** For tasks with latent pathway structure, the broadcast becomes a role vector inferred from the router. Local pathway activity gates this vector. Upward block transport then passes it to earlier branches, aligning their feedback with downstream descendants.

**Router-derived branch roles.** Each branch receives a role profile inferred from router assignments and incoming block weights rather than from a hand-coded apical/basal label. Plasticity is then modulated by branch selectivity and a synapse-specific role-alignment factor, emphasizing pathway-consistent updates without imposing a heuristic branch taxonomy.

### C.3 HSIC Auxiliary Objectives

Following [24], the figure-ground MNIST LocalCA runs optionally apply an HSIC-style auxiliary gradient to selected layer representations  $\mathbf{Z}$ :

$$\mathcal{L}^{\text{self}} = B^{-2} \text{tr}(\mathbf{K}_Z \mathbf{H} \mathbf{K}_Z \mathbf{H}), \quad \mathcal{L}^{\text{target}} = -B^{-2} \text{tr}(\mathbf{K}_Z \mathbf{H} \mathbf{K}_Y \mathbf{H}).$$

Here  $B$  is batch size,  $\mathbf{H}$  is the centering matrix,  $\mathbf{K}_Z$  uses the configured representation kernel, and  $\mathbf{K}_Y$  uses one-hot class labels when `target_source=labels`. The reported figure-ground runs use an RBF kernel with fixed bandwidth  $\sigma = 1.0$ , HSIC weight 0.01, self and target weights 0.3, five-epoch warmup, and gradient clipping at 0.1; the code applies the resulting auxiliary gradient through the configured local-learning representation pathway rather than through the conductance-factorization theorem. Moderate weights (0.01–0.1) help on figure-ground MNIST; negligible on MNIST.

## D Task Construction and Robustness Protocols

This section specifies the benchmark, synthetic-task, depth, and feedback-noise protocols used in the experiments, with emphasis on which inputs preserve the nonnegative-drive regime assumed by the conductance model.

### D.1 General Data Handling

MNIST, Fashion-MNIST, and CIFAR-10 are loaded from standard public dataset loaders, flattened before entering the dendritic core, scaled to  $[0, 1]$ , and passed through the configured nonnegative transfer stage in the main experiments. MNIST-derived synthetic tasks preserve the flattened  $28 \times 28$  geometry. Fixed random seeds define task generators such as corruption projections and context masks; generator parameters are shared across train, validation, and test splits, while sample-level corruptions are drawn per example. Validation accuracy selects checkpoints, and test accuracy is reported. CIFAR-10 is used as a flattened harder-dataset stress test under the same conductance constraints rather than as a competitive vision benchmark; no data augmentation is used as a performance device.

### D.2 Supervised Benchmark Tasks

**MNIST and Fashion-MNIST.** These ten-class tasks use nonnegative pixel inputs. MNIST tests whether per-soma 5F LocalCA approaches matched backpropagation on clean supervised data; Fashion-MNIST adds a same-dimensional distribution shift.

**CIFAR-10 harder-dataset stress test.** CIFAR-10 inputs are flattened RGB images in  $[0, 1]$ . The compact CIFAR family uses a smaller dendritic core, nonlinear decoder, and decoder-aware LocalCA mapping to test whether the broadcast-fidelity ladder persists on harder data: per-soma feedback is weak, higher-rank feedback helps, and exact effective path transport approaches the matched shunting backpropagation reference.

**Figure-ground MNIST.** For each reshaped  $28 \times 28$  MNIST image, the right half carries the digit and the left half is replaced by independent distractors  $u_{r,c} \sim \text{Uniform}(0, 0.25)$ , clipped to  $[0, 1]$  when needed. The task tests whether local credit assignment exploits spatial signal/distractor structure without leaving the nonnegative-drive regime. The main LocalCA performance rows use the HSIC auxiliary objective with weight 0.01, which adds roughly 3 pp in the ablation; the matched shunting BP reference in Table S4 is the standard cross-entropy reference.

**Noise resilience.** Flattened MNIST images are corrupted as  $\tilde{x} = \text{clip}(x + \sigma_{\text{task}}Az, 0, 1)$  with  $\sigma_{\text{task}} = 1.5$ , a fixed task-seeded projection  $A \in \mathbb{R}^{784 \times 50}$ , and fresh  $z \sim \mathcal{N}(0, I_{50})$  for each corrupted example. The fixed projection creates correlated interference without reducing the task to memorizing one corruption pattern.

**Cue integration.** Two noisy cue streams ( $A$  and  $B$ ) are presented simultaneously for two-class classification, with a one-hot context indicating which cue is reliable on each trial. Cue vectors are clipped to  $[0, 1]$  after noise. The fixed-pathway variant duplicates context into both cue branches; the learned-routing variant requires the router to discover cue separation from data. This feedback-rank diagnostic tests structured pathway-vector broadcast inside the nonnegative-input regime.

### D.3 Depth Scaling and Noise Robustness

These stress diagnostics keep the same broad model family while changing dendritic depth or corrupting the broadcast signal. For depth scaling, dendritic depth varies from 1–4 layers (branch factors [9] to [3, 3, 3, 3]): shunting local degrades from 63.5% to 57.4%, additive local falls from 54.9% to 29.7%, the shunting advantage grows from +8.5 to +27.7 pp, and matched backpropagation references remain near 90–92% (Fig. S6A). For noise robustness, Gaussian noise  $\mathcal{N}(0, \sigma^2)$  is added to the broadcast error; shunting remains robust to  $\sigma \leq 0.1$  (about 62%), while additive drops from 46.5% to chance at  $\sigma=1.0$ , indicating that shunting credit signals carry useful learning information beyond broadcast magnitude alone (Fig. S6B).